SAML

Stuttering Annotation Markup Language

W3C SSML Extension

A framework for annotating stuttering patterns
in speech data for ASR and TTS applications

Clinical-Grade Speech Annotation for Research & Industry

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Enabling accurate annotation of stuttered speech for machine learning

Bridging Clinical Research and AI Development

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Abstract

Accurate annotation of stuttered speech is essential for developing robust automatic speech recognition (ASR) systems and naturalistic text-to-speech (TTS) applications for clinical use. However, no standardized markup language currently exists for representing the complex temporal and phonetic characteristics of stuttering dysfluencies. This paper presents the **Stuttering Annotation Markup Language (SAML)**, a W3C Speech Synthesis Markup Language (SSML) extension designed to address this gap. SAML introduces XML elements and attributes for marking three primary dysfluency types (repetitions, prolongations, and blocks) with duration-based severity ratings on a clinically-motivated 0–8 scale compatible with established assessment tools such as the SSI-4. The specification supports dual functionality: annotation of recorded speech samples for ASR training corpora, and control of synthesized stuttered speech for TTS-based clinical applications. We provide a complete XML Schema Definition (XSD), comprehensive annotation guidelines with decision flowcharts for classification, and a shorthand notation system for efficient annotator workflows. The framework is designed to be language-agnostic, extensible, and fully backward-compatible with existing SSML processors. This specification aims to establish a foundation for standardized stuttered speech corpora, enabling reproducible research and improved assistive technologies for the nearly 80 million people worldwide who stutter.

Keywords: speech annotation, stuttering, dysfluency, SSML, XML schema, automatic speech recognition, text-to-speech synthesis, clinical speech analysis, severity rating, speech corpus

1 Introduction

1.1 Motivation: Bridging the AI Accessibility Gap

The Scale of Stuttering: Nearly **80 million people worldwide**, approximately 1% of the global population, stutter, with males affected two to five times more often than females. While 5–7% of children experience developmental stuttering, 75–80% recover naturally by late childhood; the remaining 15–20% develop persistent stuttering that continues into adulthood. Persistent stuttering often leads to significant mental health consequences, including social isolation, anxiety disorders, and in severe cases, suicidal ideation.

The AI "Haves" and "Have-Nots": Despite this prevalence, people who stutter (PWS) remain largely invisible to modern AI systems, creating a growing divide between those who benefit from voice technology and those excluded by it. Current automatic speech recognition systems are trained predominantly on fluent speech. When encountering stuttered speech patterns (repetitions, prolongations, and blocks), these systems fail dramatically, producing garbled transcriptions, dropped words, or complete recognition failures. This excludes PWS from voice-controlled devices, dictation software, voice assistants, voice banking, and the increasingly voice-first digital world. In an era where voice interfaces are becoming the primary mode of human-computer interaction, this exclusion compounds existing barriers faced by PWS.

The Clinical Supply-Demand Gap: There is a severe global shortage of speech-language pathologists (SLPs) specializing in fluency disorders. In many regions, wait times for stuttering therapy exceed 12–18 months, and rural or underserved communities may have no access at all. The World Health Organization estimates significant unmet need for speech therapy services globally, particularly in low- and middle-income countries. AI-powered assessment and therapy tools could help bridge this gap by enabling remote screening, automated severity measurement, and scalable intervention support, but only if we first build the annotated speech corpora needed to train them.

The Missing Foundation: To train ASR systems that understand stuttered speech, and TTS systems that can produce naturalistic stuttered speech for therapy applications, we need large-scale annotated corpora. Yet no standardized annotation format exists for stuttered speech, leading to:

- **Fragmented research**: Each research lab develops proprietary annotation schemes
- **Non-reproducible results**: Studies cannot be compared, validated, or combined
- **Wasted effort**: Annotators must be retrained for each new project
- **Limited data**: Small, incompatible datasets cannot be pooled into larger corpora

SAML addresses this gap by providing a standardized, open specification for annotating stuttered speech. With SAML, the research community can build interoperable corpora, train inclusive AI systems, and develop scalable clinical tools, ultimately working toward a world where AI works *for* people who stutter, not against them.

1.2 Purpose and Scope

This document defines SAML (Stuttering Annotation Markup Language), an extension of the W3C Speech Synthesis Markup Language (SSML) 1.1. SAML provides a standardized way to:

1. **Annotate recorded speech**: Mark dysfluencies in audio transcripts for training automatic speech recognition (ASR) systems
2. **Control speech synthesis**: Direct text-to-speech (TTS) engines to produce naturalistic stuttered speech for clinical and research applications
3. **Support clinical assessment**: Enable automated calculation of stuttering metrics compatible with established tools like SSI-4

1.3 Why Extend SSML?

SSML is a W3C recommendation that provides a standard way to control speech synthesis. By extending SSML, SAML offers several advantages:

- **Compatibility**: Works with existing SSML parsers and TTS engines
- **Standardization**: Builds on an established, well-documented specification
- **Validation**: XML Schema enables automatic validation of annotations
- **Tooling**: Leverages existing XML processing tools and libraries
- **Extensibility**: Allows future additions without breaking compatibility

1.4 Document Structure

This document is organized as follows:

Section	Content
1	Introduction and background
2	Quick start for annotators (cheat sheets, common decisions)
3	SSML basics and SAML integration
4	Namespace and schema specification
5	SAML element reference
6	Dysfluency annotation guidelines
7	Severity measurement protocol
8	Distinguishing stuttering from typical disfluencies
9	Transcription conventions
10	Special situations
11	Shorthand notation for annotators
12	Quality assurance
A–K	Appendices (ASR/TTS consumption, samples, conversion reference, element hierarchy, glossary, quick reference, severity scale, changelog, practice exercises, FAQ, tool guides)

1.5 Intended Audience

- **Data annotators**: Primary users who transcribe and mark stuttered speech
- **Software developers**: Building ASR/TTS systems with stuttering support
- **Researchers**: Studying stuttering patterns using annotated corpora
- **Clinicians**: Using annotated data for assessment and therapy

1.5.1 Quick Reference for Annotators

For annotators seeking specific information, the following sections are most relevant:

Topic	Location
Quick Start (overview, cheat sheets, common decisions)	**Section 2: Quick Start for Annotators**
How to tag repetitions, prolongations, blocks	Section 6 (Dysfluency Annotation Guidelines)
Severity ratings (0–8 scale)	Section 7 and Appendix G
What NOT to tag (interjections, phrase repetitions)	Section 8 (Distinguishing Stuttering from Typical Disfluencies)
Filler word spellings (um, uh, etc.)	Section 9.4 (Filler Words)
Shorthand notation	Section 11
Self-review checklist	Section 12.3
Comprehensive interjection list	**Appendix E: Detailed Interjection Reference**
Non-speech sounds guide	**Appendix E: Detailed Non-Speech Sound Reference**
Background sounds annotation	**Section 10.5: Background Sound Annotation**

1.5.2 Language Scope and Internationalization

This version of the SAML specification is designed primarily for **American English**. The following language-specific assumptions are made throughout this document:

- Graphemic representations follow American English spelling conventions
- Syllabification rules assume English phonotactics
- Filler word spellings (um, uh, er) reflect English usage
- Examples and shorthand notation use English vocabulary

Adapting SAML for Other Languages:

The core SAML framework (elements, attributes, severity scale) is language-agnostic and can be adapted for other languages. Implementations for non-English languages should:

1. Document language-specific grapheme conventions
2. Provide appropriate filler word inventories for the target language
3. Specify syllabification resources appropriate to the language
4. Consider language-specific dysfluency patterns (e.g., particle repetitions in Japanese, prefix repetitions in German)
5. Use the `xml:lang` attribute to indicate the document language

Future versions of this specification may include language-specific annexes. Contributors interested in developing language-specific extensions should contact the specification maintainers.

1.6 How to Read This Document

This document is comprehensive and may seem overwhelming at first. Different readers have different needs. This section provides recommended reading paths based on your role and goals.

1.6.1 Reader Profiles and Recommended Paths

Table 1: Recommended Reading Paths by Reader Profile

Profile	Primary Goal	Recommended Path
New Annotator	Start annotating quickly	1. Section 2 (Quick Start), *essential* 2. Section 6 (Dysfluency Guidelines) 3. Section 7 (Severity Measurement) 4. Section 11 (Shorthand Notation)
Experienced Annotator	Reference during work	1. Section 2 (Quick Start), bookmark this 2. Section 8 (as needed) 3. Appendix E (Glossary)
Software Developer	Implement SAML parser/writer	1. Section 4 (Schema Specification) 2. Section 5 (Element Reference) 3. Appendix A (ASR/TTS Patterns) 4. Appendix D (Element Hierarchy)
Researcher	Understand annotation decisions	1. Full document read recommended 2. Focus on Section 8 for edge cases 3. Section 12 for reliability metrics
Project Manager	Set up annotation project	1. Section 1 (Introduction) 2. Section 12 (Quality Assurance) 3. Subsection 12.5 (Inter-Rater Reliability)

1.6.2 Section Dependencies

Some sections build on concepts introduced earlier. Use this dependency guide:

- **Before reading Section 6** (Dysfluency Guidelines): Read Section 2 to understand the basics
- **Before reading Section 11** (Shorthand Notation): Read Section 6 to understand what you're annotating
- **Before reading Appendix A** (ASR/TTS Patterns): Read Section 4 for schema understanding
- **Section 3** (SSML Basics): Optional unless you're building software or need XML details

1.6.3 Quick Start by Goal

Table 2: Start Here Based on Your Goal

"I want to..."	Start Here
Annotate my first audio file	Section 2
Look up a specific dysfluency type	Section 6
Decide if something is stuttering	Section 8
Know what severity rating to use	Section 7
Use shorthand instead of XML	Section 11
Build a SAML parser	Section 4
Set up quality control	Section 12
Find common interjections	Appendix E

Note

Tip: Print or bookmark Section 2 (Quick Start for Annotators). It contains cheat sheets and quick references you'll use repeatedly during annotation work.

1.7 Example Audio and Training Resources

Annotators benefit from practicing with audio examples before working on project data. This section provides guidance on finding and using training materials.

1.7.1 Recommended Public Datasets

The following publicly available datasets contain stuttered speech suitable for practice:

Table 3: Public Stuttered Speech Datasets

Dataset	Access	Notes
UCLASS (UCL Archive of Stuttered Speech)	Request from UCL	British English; multiple speakers; varied severity
FluencyBank	`fluency.talkbank.org`	Part of TalkBank; requires registration; includes transcripts
SEP-28k	Academic request	28,000 clips; crowd-sourced annotations; podcasts
LibriStutter	GitHub/Academic	Synthetic stuttered speech; useful for baseline practice

Note

Licensing: Always check dataset licenses before use. Most academic datasets require signing a data use agreement and prohibit redistribution. Do not share downloaded samples outside your project team.

1.7.2 Using Practice Materials

When using practice audio:

1. **Start with clear examples**: Begin with recordings that have obvious, well-separated dysfluencies
2. **Progress to complex cases**: Move to recordings with rapid speech, multiple dysfluency types, or background noise
3. **Compare with gold standard**: If the dataset includes annotations, compare your annotations against them
4. **Discuss disagreements**: When your annotations differ from gold standard, discuss with supervisors to understand the reasoning

1.7.3 Creating Internal Training Sets

For project-specific training:

1. **Select representative samples**: Choose 10–20 audio clips that cover all dysfluency types (R, P, B) and severity levels (0–8)
2. **Create gold annotations**: Have senior annotators create consensus annotations
3. **Document edge cases**: Note any ambiguous examples and the reasoning for classification decisions
4. **Update regularly**: Add new edge cases as they arise during production annotation

1.7.4 Requesting Project-Specific Training Materials

If your organization maintains a repository of training materials:

- Contact your project lead or annotation manager
- Request access to the internal training set
- Complete any required onboarding before accessing sensitive data
- Do not share training materials outside the project

Note

See Also: Appendix I (Practice Exercises) contains written exercises with answer keys that do not require audio files. These are useful for learning classification rules before working with actual recordings.

1.8 Audio Preparation and Chunking Guidelines

1.8.1 Why Chunking is Required

Raw audio recordings from clinical sessions, interviews, or conversations are often long, ranging from several minutes to an hour or more. However, machine learning models for ASR (Automatic Speech Recognition) and TTS (Text-to-Speech) cannot be trained on such long audio files directly.

Chunking solves this problem by:

- **Creating training samples**: Each chunk becomes one training sample paired with its transcript
- **Enabling batch processing**: Short, uniform-length segments can be batched efficiently during training
- **Managing memory**: Neural networks have memory constraints; shorter segments fit within model context windows
- **Improving alignment**: Shorter segments allow for more accurate audio-text alignment

Chunking a Long Recording

A 20-minute clinical interview recording would be segmented into approximately 100–120 chunks of 10 seconds each. Each chunk is then transcribed and annotated separately, creating 100–120 training samples from a single recording session.

1.8.2 Chunk Duration Requirements

The target chunk duration balances two competing needs:

- **Short enough**: To fit model requirements and enable efficient training
- **Long enough**: To contain complete utterances and meaningful linguistic units

Table 4: Chunk Duration Specifications

Parameter	Value
Target mean duration	10 seconds
Standard deviation	2 seconds
Start looking for boundaries	6 seconds (mean − 2 SD)
Ideal range	8–12 seconds
Acceptable range	5–15 seconds
Absolute maximum (force break)	15 seconds

1.8.3 When to Create a Chunk Boundary

Create a new chunk (segment break) when:

1. The current segment reaches 6+ seconds AND a natural pause occurs
2. The speaker completes a thought or sentence with a clear pause
3. A different speaker begins talking (in multi-speaker audio)
4. There is a significant topic change with a natural break
5. The speaker takes a breath pause of 500ms or longer at a grammatical boundary

Good Chunk Boundaries

- End of a sentence with falling intonation
- Between speaker turns in a conversation
- After a complete thought: "I went to the store yesterday." [CHUNK] "Then I came home."
- At paragraph boundaries in reading tasks

1.8.4 When NOT to Create a Chunk Boundary

Never create a chunk boundary:

1. **In the middle of a dysfluency event**: If a block, prolongation, or repetition is occurring, extend the chunk to include the complete event
2. **In the middle of a word**: Even if the segment is getting long
3. **In the middle of a phrase**: Keep syntactic units together
4. **During an incomplete thought**: Wait for the speaker to finish

5. **At a brief hesitation pause**: Short pauses within sentences are not chunk boundaries
6. **Simply because you reached 10 seconds**: Natural boundaries take priority over duration targets; start looking for boundaries at 6 seconds but don't force them

Warning

Critical Rule: Never cut a segment in the middle of a dysfluency event. If a block or prolongation extends past your target endpoint, you must extend the segment to include the complete dysfluency and the target word that follows.

× **Incorrect**

Wrong: Cutting during a block

```
Chunk 1: I want to ( [IBC 2]
Chunk 2: ) go to the store.
```

The block is split across chunks, making annotation invalid.

✓ **Correct**

Correct: Keeping the dysfluency intact

```
Chunk 1: I want to ( [IBC 2] ) go to the store.
Chunk 2: Maybe we can get some milk.
```

The entire dysfluency event is contained in one chunk.

1.8.5 Handling Long Sentences

When a speaker produces a very long sentence without natural pauses:

1. **Do not artificially break the sentence**: Keep it in one chunk even if it exceeds 12 seconds
2. **Exception**: If the chunk would exceed 15 seconds, find the least disruptive point (e.g., a conjunction like "and," "but," "so")
3. **Never break within a clause**: Keep subject-verb-object units together

Handling a 25-second Sentence

Original: "I went to the store and I bought some milk and bread and eggs and then I went to the pharmacy to pick up my prescription and after that I stopped at the bank."

Acceptable break point: "...to pick up my prescription" [CHUNK] "and after that I stopped at the bank."

Unacceptable break point: "...to pick up my" [CHUNK] "prescription and after..."

1.8.6 Multi-Speaker Chunking

When audio contains multiple speakers:

1. Create a new chunk/segment for each speaker turn
2. Even single-word responses get their own segment
3. Label each chunk with the speaker identifier
4. Overlapping speech stays with the primary speaker's chunk (use `<overlap>` tag)

Multi-Speaker Segmentation

```
[Chunk 1 - Speaker 1]: Hello, how are you doing today?
[Chunk 2 - Speaker 2]: ( [R:syl 1] I- I- ) I'm doing well, thanks.
[Chunk 3 - Speaker 1]: Great!
```

```
[Chunk 4 - Speaker 2]: Yeah.
[Chunk 5 - Speaker 1]: So, what brings you in today?
```

Note: Even "Great!" and "Yeah" are separate chunks. The "I-I-" repetition uses `:syl` because "I" is a monosyllabic word (syllable repetition).

1.8.7 Chunking Decision Process

Follow these steps in order to decide when to create a chunk boundary:

Table 5: Chunking Decision Steps

Step	Question	Action
1	Is current duration < 6 seconds?	**Yes**: Continue listening, do not chunk yet **No**: Go to Step 2
2	Is there a natural pause or speaker change?	**No**: Go to Step 3 **Yes**: Go to Step 4
3	Is current duration > 15 seconds?	**No**: Continue, wait for natural pause **Yes**: Find nearest phrase boundary and create chunk
4	Are we inside a dysfluency event?	**No**: **Create chunk boundary here** **Yes**: Wait for dysfluency to complete, then create chunk

Chunking Decision Examples

Situation	Duration	Decision
Speaker is mid-sentence	4 sec	Continue (under 6 sec)
Speaker pauses at sentence end	8 sec	Chunk here (natural pause, not in dysfluency)
Speaker pauses during a block	9 sec	Wait for block to end, then chunk
No pause, continuous speech	12 sec	Continue waiting for natural pause
No pause, continuous speech	16 sec	Force chunk at nearest phrase boundary
Speaker change (new person)	7 sec	Chunk here (speaker change)

Note

Golden Rule: Never break inside a dysfluency. If you reach 15 seconds while inside a prolongation, block, or repetition, wait for it to complete before chunking.

1.8.8 Audio Technical Requirements

Table 6: Audio File Specifications

Parameter	Requirement
Format	WAV (preferred), FLAC, or MP3
Sample rate	Minimum 16 kHz (44.1 kHz recommended)
Bit depth	Minimum 16-bit
Channels	Mono preferred; stereo acceptable

2
Quick Start for Annotators

This section provides a condensed reference for annotators who need to begin annotating quickly or need a quick lookup during annotation work. For full details, refer to the sections referenced in each subsection.

2.1 The 5-Minute Overview

What is SAML?
SAML (Stuttering Annotation Markup Language) is an XML-based format for annotating stuttered speech. It extends SSML (Speech Synthesis Markup Language) with elements specifically designed to capture dysfluency events.

What do annotators do?
Annotators listen to audio recordings and mark three types of dysfluency events:

1. **Repetitions**: When sounds, syllables, or monosyllabic words are repeated ("b-b-baby", "I-I-I")
2. **Prolongations**: When sounds are stretched beyond normal duration ("sssssnake")
3. **Blocks**: When speech stops involuntarily before or during a word (silent struggle or audible tension)

Core workflow:

1. Listen to an audio chunk (typically 5–15 seconds)
2. Identify any dysfluency events
3. For each event: determine type, measure duration, assign severity
4. Transcribe using shorthand notation or full XML

Key principle: Only tag *involuntary* dysfluencies characteristic of stuttering. Do not tag normal hesitations, filler words (um, uh), or intentional repetitions for emphasis.

2.1.1 Overall Annotation Workflow

This diagram shows the complete annotation process from audio file to final SAML output:

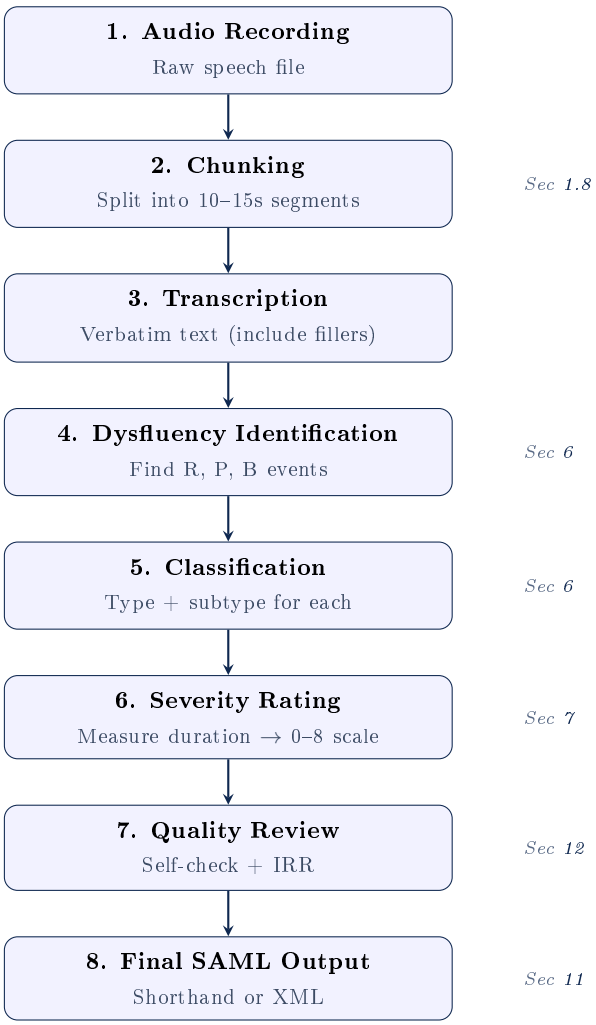

Figure 1: SAML Annotation Workflow

2.2 Severity Scale Quick Reference

All dysfluency events require a severity rating (0–8) based on duration. Use this table for quick lookup:

Table 7: Severity Scale Quick Reference

Rating	Duration	Remember As
0	< 0.5s	Half second or less
1	0.5–1s	Under one second
2	1–2s	One to two seconds
3	2–3s	Two to three seconds
4	3–5s	Three to five seconds
5	5–10s	Five to ten seconds
6	10–30s	Ten to thirty seconds
7	30–60s	Half minute to one minute
8	≥ 60s	One minute or more

Note

Tip: Boundary values go to the higher rating. Exactly 1.0 seconds = rating 2, not 1.

2.3 Tag / Don't Tag Quick Reference

Use this table to quickly decide whether to tag something. For the complete decision summary, see Table 25.

Table 8: Tag / Don't Tag Quick Reference

What You Hear	Tag?	Type
Repetitions		
Sound/partial syllable repeated (b-b-baby)	✓	R:isyl
Complete syllable repeated (ba-ba-baby)	✓	R:syl
Monosyllabic word repeated (I-I-I)	✓	R:syl
Mixed ISR+SR (t-t-to-to-toffee)	✓	R:mix
Multi-syllable word repeated (maybe-maybe)	×	N/A
Phrase repeated (I want to-I want to)	×	N/A
Prolongations		
Sound stretched involuntarily (ssss-nake)	✓	P
Sound stretched for emphasis (sooooo good)	×	N/A
Blocks		
Speech stops, visible/audible struggle	✓	B/IBC/ABC
Natural pause between thoughts	×	N/A
Other		
Um, uh, like, you know	×	N/A
False start / revision (I went-I drove)	×	N/A
Intentional repetition for emphasis	×	N/A

2.4 Shorthand Cheat Sheet

Shorthand notation is faster than full XML. Here's the complete reference:

Table 9: Shorthand Notation Cheat Sheet

Type	Shorthand	Example
Repetitions		
Incomplete syllable (ISR)	`( [R:isyl N] ... )`	`( [R:isyl 1] b- b- ) baby`
Syllable (SR)	`( [R:syl N] ... )`	`( [R:syl 1] ba- ba- ) baby`
Mixed	`( [R:mix N] ... )`	`( [R:mix 2] t- t- to- ) toffee`
Prolongations		
Prolonged sound	`( [P N] X- )rest`	`s( [P 2] s- )ee`
Blocks		
Inaudible consonant	`( [IBC N] )`	`( [IBC 2] ) baby`
Audible consonant	`( [ABC N] )`	`( [ABC 1] ) go`
Inaudible vowel	`( [IBV N] )`	`( [IBV 1] ) apple`
Audible vowel	`( [ABV N] )`	`( [ABV 2] ) every`
Generic block	`( [B N] )`	`( [B 1] ) word`
Other		
Unintelligible (guess)	`{g: word}`	`{g: probably}`
Unintelligible (count)	`{w: N}`	`{w: 3}`
Non-speech sound	`[type]`	`[laughs] [coughs]`
Background sound	`[bg:type level]`	`[bg:traffic] [bg:music soft]`
Letter spelling	`~X`	`~A ~B ~C`

Key syntax rules:

- `N` = severity rating (0–8)
- Parentheses `( )` enclose dysfluency content
- Square brackets `[ ]` contain tag type and severity
- Add hyphen after each repeated/prolonged unit
- Block tags have empty content between parentheses

2.5 Most Common Annotation Decisions

These are the situations annotators encounter most frequently:

Table 10: Most Common Annotation Scenarios

Scenario	Decision	Reasoning
"I-I-I want to go"	Tag as R:syl	"I" is a monosyllabic word
"um, I think that..."	Don't tag	"Um" is a filler, not stuttering
"b-b-baby"	Tag as R:isyl	"b" is not a complete syllable
"ba-ba-baby"	Tag as R:syl	"ba" is a complete syllable of "baby"
"d-d-do-do-dog"	Tag as R:isyl	All units are incomplete; varying lengths still = ISR
"t-t-to-to-toffee"	Tag as R:mix	Has both "t" (incomplete) AND "to" (complete syllable)
"ssssnake" (involuntary)	Tag as P	Sound prolonged due to stuttering
"nooooo way!" (emphasis)	Don't tag	Intentional prolongation for effect
(silence, then) "...baby" with struggle	Tag as block	Involuntary stoppage before word
"I went... I mean I drove"	Don't tag	This is a revision/false start
"maybe-maybe I should"	Don't tag	Multi-syllable word repetition
"I, I was thinking" (deliberate)	Don't tag	Intentional repetition for thinking

Note

When in doubt, ask: "Did this happen involuntarily? Does it look/sound like the speaker struggled or lost control?" If yes, it's likely stuttering and should be tagged. If it seems deliberate or effortless, don't tag it.

2.6 Common Mistakes and Pitfalls

These are the most frequent errors new annotators make. Review this list before starting and revisit it periodically.

Table 11: Top 10 Annotation Pitfalls

#	Mistake	Correct Approach
1	Tagging "um", "uh", "like"	These are normal fillers. Write them in transcript but do NOT tag.
2	Confusing thinking pauses with blocks	Blocks show *struggle or tension*. Simple pauses to think are not blocks.
3	Classifying "I-I-I" as ISR	Monosyllabic word repetitions are **SR** (syllable), not ISR. "I" = complete syllable.
4	Tagging "maybe-maybe"	Multi-syllable word repetitions are normal disfluency. Do NOT tag.
5	Using Mixed for varying ISR lengths	"d-d-do-do-dog" with varying lengths is still **ISR**. Mixed requires both ISR AND SR units.
6	Tagging emphatic prolongations	"Sooooo good!" with emphasis is intentional. Only tag *involuntary* prolongations.
7	Block type based on spelling	Use *phonetic onset*. "honest" starts with vowel /o/, not consonant "h".
8	Including target word in duration	Measure only the repeated/prolonged units, **excluding** the final target word.
9	Tagging self-corrections	"I went... I mean I drove" is a revision, not stuttering. Do NOT tag.
10	Breaking chunks mid-dysfluency	Never cut audio chunks during a dysfluency. Extend to include complete event.

Warning

Most Common Error: Tagging normal disfluencies as stuttering. Remember: fillers (um, uh), phrase repetitions, multi-syllable word repetitions, and self-corrections are **normal speech patterns** that everyone produces. Only tag *involuntary* repetitions, prolongations, and blocks.

Wrong vs. Right Examples

Wrong	Right
`( [R:isyl 1] I- I- ) I want` *"I" is monosyllabic = SR, not ISR*	`( [R:syl 1] I- I- ) I want`
`( [R:syl 1] um- um- ) um` *Don't tag fillers*	`um, I want to go`
`( [IBC 2] ) honest` (consonant block) *"h" is silent; phonetic onset is /o/*	`( [IBV 2] ) honest` (vowel block)
`( [R:mix 1] d- d- do- do- ) dog` *All units are incomplete = ISR*	`( [R:isyl 1] d- d- do- do- ) dog`

3 SSML Basics and SAML Integration

3.1 Overview of SSML

The Speech Synthesis Markup Language (SSML) is an XML-based markup language defined by the W3C. It controls how text is converted to speech by TTS engines. Key SSML elements include:

Table 12: Core SSML Elements

Element	Purpose	Example
`<speak>`	Root element	Required container for all content
`<voice>`	Voice selection	Change speaker voice
`<prosody>`	Speech properties	Control rate, pitch, volume
`<break>`	Pause insertion	Add silence of specified duration
`<emphasis>`	Stress words	Add emphasis to content
`<phoneme>`	Pronunciation	Specify exact phonetic rendering
`<sub>`	Substitution	Replace text with spoken form
`<say-as>`	Interpretation	Specify how to interpret content

3.2 Basic SSML Structure

A minimal SSML document looks like this:

```
<!-- Minimal SSML document -->
<?xml version="1.0" encoding="UTF-8"?>
<speak version="1.1"
       xmlns="http://www.w3.org/2001/10/synthesis"
       xml:lang="en-US">
    Hello, this is a simple SSML document.
</speak>
```

3.3 How SAML Extends SSML

SAML adds new elements within a separate XML namespace. This means:

- SAML elements are prefixed with `saml:`
- Standard SSML elements work normally alongside SAML elements
- Parsers that don't understand SAML can ignore the extensions gracefully

3.4 Basic SAML Document Structure

A SAML-annotated document includes both namespaces:

```
<!-- SAML document with namespace declarations -->
<?xml version="1.0" encoding="UTF-8"?>
<speak version="1.1"
       xmlns="http://www.w3.org/2001/10/synthesis"
       xmlns:saml="https://shunyalabs.ai/2026/saml"
       xml:lang="en-US">

    <!-- Syllable repetition example: "ba-ba-baby" -->
    I want a <saml:repetition type="syllable" severity="1">
        <saml:unit>ba</saml:unit>
        <saml:unit>ba</saml:unit>
    </saml:repetition> baby.

</speak>
```

3.5 SAML and SSML Interaction

SAML elements can be combined with standard SSML elements:

Note

SAML-Specific Extensions: The following SAML elements are **custom extensions** to SSML and are **not** part of the W3C SSML standard:

- `<saml:repetition>`, `<saml:prolongation>`, `<saml:block>`: dysfluency elements
- `<saml:nonspeech>`: non-speech sound annotation (SSML uses `<audio>` for sound files)
- `<saml:unintelligible>`, `<saml:overlap>`, `<saml:letter>`: auxiliary elements
- `<saml:metadata>`: file and speaker metadata

TTS engines that do not support SAML extensions should gracefully ignore these elements. For TTS synthesis of non-speech sounds, implementations may need to use SSML's `<audio>` element with sound files as a fallback.

```
<!-- Combining SAML with SSML prosody control -->
<speak version="1.1"
       xmlns="http://www.w3.org/2001/10/synthesis"
       xmlns:saml="https://shunyalabs.ai/2026/saml"
       xml:lang="en-US">

    <prosody rate="slow">
        <!-- Block before "think" -->
        I <saml:block type="audible" sound="consonant" severity="2"/>
        think we should go.
    </prosody>

</speak>
```

4

Namespace and Schema Specification

4.1 Recommended Namespace Declaration

We recommend that implementations of SAML adopt a namespace URI following this pattern:

`https://[organization-domain]/[year]/saml`

For example, an organization implementing SAML might use:

- `https://example-university.edu/2026/saml`
- `https://speech-research-org.org/2026/saml`

The recommended namespace prefix is `saml`. All SAML elements and attributes should be qualified with this namespace.

Note

Once a formal namespace is established by the community or a standards body, this document will be updated to reflect the canonical URI.

4.2 Recommended Schema Location

Implementations should host the XML Schema Definition (XSD) at a stable URL and reference it in documents. For example:

`https://[organization-domain]/[year]/saml/saml.xsd`

4.3 Namespace Governance and Versioning

To prevent fragmentation and ensure interoperability across implementations, the following governance policies are recommended:

4.3.1 Canonical Namespace

The canonical namespace for SAML version 1.0 is:

`https://shunyalabs.ai/2026/saml`

This namespace is officially hosted by Shunya Labs and serves as the authoritative reference for SAML implementations. The XML Schema Definition (XSD) is available at:

`https://shunyalabs.ai/2026/saml/saml.xsd`

Organizations creating compatible implementations MUST use this namespace to ensure interoperability. Custom namespaces should only be used for experimental or incompatible extensions that do not aim for cross-implementation compatibility.

Note

The specification, schema, and tools are available at `https://shunyalabs.ai/saml`. This includes the complete XML Schema Definition, the shorthand-to-XML converter, and sample annotations.

4.3.2 Versioning Policy

Table 13: Versioning Policy

Version Type	Policy
Patch (1.0.x)	Bug fixes, clarifications, editorial changes. No schema changes. Full backward compatibility.
Minor (1.x.0)	New optional elements/attributes. Existing elements unchanged. Documents valid under 1.0 remain valid under 1.x.
Major (x.0.0)	Breaking changes permitted. New namespace URI required (e.g., `.../2026/saml`). Migration guide must be provided.

4.3.3 Backward Compatibility Guarantees

- **Minor versions**: Documents valid under version 1.0 MUST validate under any 1.x schema
- **Element semantics**: The meaning of existing elements and attributes MUST NOT change within a major version
- **Deprecation**: Elements may be deprecated (with warnings) but not removed within a major version
- **New requirements**: New required attributes MUST NOT be added to existing elements in minor versions

4.4 Document Declaration

A complete SAML document should include declarations following this pattern:

```
<?xml version="1.0" encoding="UTF-8"?>
<speak version="1.1"
       xmlns="http://www.w3.org/2001/10/synthesis"
       xmlns:saml="https://shunyalabs.ai/2026/saml"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="https://shunyalabs.ai/2026/saml
                           https://shunyalabs.ai/2026/saml/saml.xsd"
       xml:lang="en-US">

    <!-- Content here -->

</speak>
```

4.5 Proposed XML Schema Definition

The following is the proposed XSD for SAML. Implementations should adapt the namespace URI as appropriate:

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema"
           xmlns:saml="https://shunyalabs.ai/2026/saml"
           targetNamespace="https://shunyalabs.ai/2026/saml"
           elementFormDefault="qualified">

    <!-- Severity type: 0-8 based on duration -->
    <xs:simpleType name="severityType">
        <xs:restriction base="xs:integer">
            <xs:minInclusive value="0"/>
            <xs:maxInclusive value="8"/>
        </xs:restriction>
    </xs:simpleType>

    <!-- Duration in milliseconds -->
    <xs:simpleType name="durationType">
        <xs:restriction base="xs:positiveInteger"/>
    </xs:simpleType>

    <!-- Repetition types -->
    <!-- Note: "sound" and "word" types have been removed per specification revision.
         Sound repetitions are now classified as incomplete-syllable (ISR).
         Monosyllabic word repetitions are now classified as syllable (SR). -->
    <xs:simpleType name="repetitionType">
        <xs:restriction base="xs:string">
            <xs:enumeration value="incomplete-syllable"/>
            <xs:enumeration value="syllable"/>
            <xs:enumeration value="mixed"/>
        </xs:restriction>
    </xs:simpleType>

    <!-- Block sound types -->
    <xs:simpleType name="blockSoundType">
        <xs:restriction base="xs:string">
            <xs:enumeration value="consonant"/>
            <xs:enumeration value="vowel"/>
        </xs:restriction>
    </xs:simpleType>

    <!-- Block audibility types -->
    <xs:simpleType name="blockAudibilityType">
```

```
        <xs:restriction base="xs:string">
            <xs:enumeration value="audible"/>
            <xs:enumeration value="inaudible"/>
        </xs:restriction>
    </xs:simpleType>

    <!-- Unit element for repeated segments -->
    <xs:element name="unit">
        <xs:complexType mixed="true">
            <xs:attribute name="phoneme" type="xs:string"
                          use="optional"/>
        </xs:complexType>
    </xs:element>

    <!-- Repetition element -->
    <!-- Note: minOccurs="2" enforces that repetitions have at least 2 units -->
    <!-- The 'count' attribute is informational and should match unit count -->
    <xs:element name="repetition">
        <xs:complexType>
            <xs:sequence>
                <xs:element ref="saml:unit" minOccurs="2" maxOccurs="unbounded"/>
            </xs:sequence>
            <xs:attribute name="type" type="saml:repetitionType"
                          use="required"/>
            <xs:attribute name="severity" type="saml:severityType"
                          use="required"/>
            <xs:attribute name="duration" type="saml:durationType"
                          use="optional"/>
            <!-- count is informational; actual unit elements are authoritative -->
            <xs:attribute name="count" type="xs:positiveInteger"
                          use="optional"/>
        </xs:complexType>
    </xs:element>

    <!-- Prolongation element -->
    <!-- Note: 'sound' attribute is authoritative for identifying prolonged grapheme -->
    <xs:element name="prolongation">
        <xs:complexType mixed="true">
            <xs:attribute name="sound" type="xs:string"
                          use="required"/>
            <xs:attribute name="severity" type="saml:severityType"
                          use="required"/>
            <!-- duration is informational; severity is normative for classification -->
            <xs:attribute name="duration" type="saml:durationType"
                          use="optional"/>
            <xs:attribute name="position" use="optional">
                <xs:simpleType>
                    <xs:restriction base="xs:string">
                        <xs:enumeration value="initial"/>
                        <xs:enumeration value="medial"/>
                        <xs:enumeration value="final"/>
                    </xs:restriction>
                </xs:simpleType>
            </xs:attribute>
        </xs:complexType>
    </xs:element>

    <!-- Block element -->
    <xs:element name="block">
        <xs:complexType>
            <xs:attribute name="type" type="saml:blockAudibilityType"
                          use="required"/>
            <xs:attribute name="sound" type="saml:blockSoundType"
                          use="required"/>
            <xs:attribute name="severity" type="saml:severityType"
                          use="required"/>
            <xs:attribute name="duration" type="saml:durationType"
                          use="optional"/>
        </xs:complexType>
    </xs:element>

    <!-- Unintelligible speech element -->
    <xs:element name="unintelligible">
        <xs:complexType>
            <xs:attribute name="type" use="required">
                <xs:simpleType>
```

```
                    <xs:restriction base="xs:string">
                        <xs:enumeration value="guess"/>
                        <xs:enumeration value="count"/>
                        <xs:enumeration value="unknown"/>
                    </xs:restriction>
                </xs:simpleType>
            </xs:attribute>
            <xs:attribute name="value" type="xs:string"
                          use="optional"/>
            <xs:attribute name="wordCount" type="xs:positiveInteger"
                          use="optional"/>
        </xs:complexType>
    </xs:element>

    <!-- Overlap marker -->
    <xs:element name="overlap">
        <xs:complexType>
            <xs:attribute name="speaker" type="xs:string"
                          use="optional"/>
        </xs:complexType>
    </xs:element>

    <!-- Spoken letter -->
    <xs:element name="letter">
        <xs:complexType mixed="true"/>
    </xs:element>

    <!-- Non-speech sound (speaker-produced) -->
    <xs:element name="nonspeech">
        <xs:complexType>
            <xs:attribute name="type" use="required">
                <xs:simpleType>
                    <xs:restriction base="xs:string">
                        <xs:enumeration value="laugh"/>
                        <xs:enumeration value="cough"/>
                        <xs:enumeration value="sigh"/>
                        <xs:enumeration value="cry"/>
                        <xs:enumeration value="clear-throat"/>
                        <xs:enumeration value="sniff"/>
                        <xs:enumeration value="inhale"/>
                        <xs:enumeration value="exhale"/>
                        <xs:enumeration value="other"/>
                    </xs:restriction>
                </xs:simpleType>
            </xs:attribute>
            <xs:attribute name="description" type="xs:string"
                          use="optional"/>
        </xs:complexType>
    </xs:element>

    <!-- Background sound (environmental/external) -->
    <xs:element name="background">
        <xs:complexType>
            <xs:attribute name="type" use="required">
                <xs:simpleType>
                    <xs:restriction base="xs:string">
                        <!-- Traffic & Transportation -->
                        <xs:enumeration value="traffic"/>
                        <xs:enumeration value="siren"/>
                        <xs:enumeration value="horn"/>
                        <xs:enumeration value="aircraft"/>
                        <xs:enumeration value="train"/>
                        <!-- Environmental -->
                        <xs:enumeration value="wind"/>
                        <xs:enumeration value="rain"/>
                        <xs:enumeration value="thunder"/>
                        <xs:enumeration value="birds"/>
                        <xs:enumeration value="animals"/>
                        <xs:enumeration value="water"/>
                        <!-- Human Activity -->
                        <xs:enumeration value="crowd"/>
                        <xs:enumeration value="conversation"/>
                        <xs:enumeration value="children"/>
                        <xs:enumeration value="footsteps"/>
                        <!-- Indoor/Household -->
                        <xs:enumeration value="hvac"/>
```

```
                        <xs:enumeration value="appliance"/>
                        <xs:enumeration value="tv"/>
                        <xs:enumeration value="music"/>
                        <xs:enumeration value="phone"/>
                        <xs:enumeration value="door"/>
                        <!-- Technical/Equipment -->
                        <xs:enumeration value="static"/>
                        <xs:enumeration value="hum"/>
                        <xs:enumeration value="echo"/>
                        <xs:enumeration value="mic-noise"/>
                        <!-- Other -->
                        <xs:enumeration value="other"/>
                    </xs:restriction>
                </xs:simpleType>
            </xs:attribute>
            <xs:attribute name="level" use="optional">
                <xs:simpleType>
                    <xs:restriction base="xs:string">
                        <xs:enumeration value="soft"/>
                        <xs:enumeration value="moderate"/>
                        <xs:enumeration value="loud"/>
                        <xs:enumeration value="overwhelming"/>
                    </xs:restriction>
                </xs:simpleType>
            </xs:attribute>
            <xs:attribute name="description" type="xs:string"
                          use="optional"/>
            <xs:attribute name="continuous" type="xs:boolean"
                          use="optional"/>
        </xs:complexType>
    </xs:element>

    <!-- Metadata element for file/speaker info -->
    <!-- Metadata is RECOMMENDED for all documents -->
    <xs:element name="metadata">
        <xs:complexType>
            <xs:sequence>
                <xs:element name="file" minOccurs="0">
                    <xs:complexType>
                        <xs:attribute name="id" type="xs:string" use="required"/>
                        <xs:attribute name="duration" type="saml:durationType"/>
                        <xs:attribute name="task" use="optional">
                            <xs:simpleType>
                                <xs:restriction base="xs:string">
                                    <xs:enumeration value="reading"/>
                                    <xs:enumeration value="monologue"/>
                                    <xs:enumeration value="conversation"/>
                                    <xs:enumeration value="phone"/>
                                    <xs:enumeration value="presentation"/>
                                    <xs:enumeration value="spontaneous"/>
                                    <xs:enumeration value="other"/>
                                </xs:restriction>
                            </xs:simpleType>
                        </xs:attribute>
                    </xs:complexType>
                </xs:element>
                <!-- Multiple speaker elements allowed for multi-speaker files -->
                <xs:element name="speaker" minOccurs="0" maxOccurs="unbounded">
                    <xs:complexType>
                        <xs:attribute name="id" type="xs:string" use="required"/>
                        <xs:attribute name="age-group" use="optional">
                            <xs:simpleType>
                                <xs:restriction base="xs:string">
                                    <xs:enumeration value="child"/>
                                    <xs:enumeration value="adolescent"/>
                                    <xs:enumeration value="adult"/>
                                </xs:restriction>
                            </xs:simpleType>
                        </xs:attribute>
                        <xs:attribute name="severity" use="optional">
                            <xs:simpleType>
                                <xs:restriction base="xs:string">
                                    <xs:enumeration value="mild"/>
                                    <xs:enumeration value="moderate"/>
                                    <xs:enumeration value="severe"/>
                                    <xs:enumeration value="very-severe"/>
```

```
                                    <xs:enumeration value="unassessed"/>
                                </xs:restriction>
                            </xs:simpleType>
                        </xs:attribute>
                        <xs:attribute name="gender" use="optional">
                            <xs:simpleType>
                                <xs:restriction base="xs:string">
                                    <xs:enumeration value="male"/>
                                    <xs:enumeration value="female"/>
                                    <xs:enumeration value="other"/>
                                    <xs:enumeration value="unspecified"/>
                                </xs:restriction>
                            </xs:simpleType>
                        </xs:attribute>
                    </xs:complexType>
                </xs:element>
            </xs:sequence>
        </xs:complexType>
    </xs:element>

</xs:schema>
```

4.6 Attribute Semantics: Duration vs. Severity

Note

Normative vs. Informational Attributes:

- **severity** (required): The **normative** classification of dysfluency duration. This is used for clinical metrics and must be present.
- **duration** (optional): An **informational** attribute recording the exact measured duration in milliseconds. Useful for research but not required for validation.
- If both are present and inconsistent, **severity is authoritative** for classification purposes.

5 SAML Element Reference

This section provides a complete reference for all SAML elements.

5.1 Dysfluency Elements

5.1.1 The `<saml:repetition>` Element

Marks repeated incomplete syllables, syllables, or combinations thereof.

Table 14: `<saml:repetition>` Attributes

Attribute	Type	Required	Description
`type`	Enumeration	Yes	`incomplete-syllable`, `syllable`, `mixed`
`severity`	Integer 0–8	Yes	Duration-based severity rating
`duration`	Integer	No	Actual duration in milliseconds
`count`	Integer	No	Number of repetition units

Child Elements: One or more `<saml:unit>` elements containing the repeated content.

```
<!-- Incomplete Syllable Repetition (ISR): "b-b-b-baby" -->
<saml:repetition type="incomplete-syllable" severity="1">
    <saml:unit>b</saml:unit>
    <saml:unit>b</saml:unit>
    <saml:unit>b</saml:unit>
</saml:repetition> baby

```

```
<!-- Incomplete Syllable Repetition with varying lengths: "d-d-do-do-dog" -->
<saml:repetition type="incomplete-syllable" severity="2">
    <saml:unit>d</saml:unit>
    <saml:unit>d</saml:unit>
    <saml:unit>do</saml:unit>
    <saml:unit>do</saml:unit>
</saml:repetition> dog

<!-- Syllable Repetition (SR): "ba-ba-baby" -->
<saml:repetition type="syllable" severity="2" count="3">
    <saml:unit>ba</saml:unit>
    <saml:unit>ba</saml:unit>
    <saml:unit>ba</saml:unit>
</saml:repetition> baby

<!-- Syllable Repetition for monosyllabic words: "dog-dog-dog" -->
<saml:repetition type="syllable" severity="1">
    <saml:unit>dog</saml:unit>
    <saml:unit>dog</saml:unit>
</saml:repetition> dog

<!-- Mixed Repetition (ISR + SR): "t-t-to-to-toffee" -->
<saml:repetition type="mixed" severity="2">
    <saml:unit>t</saml:unit>
    <saml:unit>t</saml:unit>
    <saml:unit>to</saml:unit>
    <saml:unit>to</saml:unit>
</saml:repetition> toffee
```

5.1.2 The `<saml:unit>` Element

Contains a single repeated segment within a repetition.

Table 15: `<saml:unit>` Attributes

Attribute	Type	Required	Description
`phoneme`	String	No	IPA representation of the sound

Content: The graphemic (written) form of the repeated unit.

```
<!-- With optional phoneme attribute -->
<saml:unit phoneme="b">b</saml:unit>
<saml:unit phoneme="ba">ba</saml:unit>
```

5.1.3 The `<saml:prolongation>` Element

Marks sounds that are stretched longer than normal.

Table 16: `<saml:prolongation>` Attributes

Attribute	Type	Required	Description
`sound`	String	Yes	The prolonged grapheme(s)
`severity`	Integer 0–8	Yes	Duration-based severity rating
`duration`	Integer	No	Actual duration in milliseconds
`position`	Enumeration	No	`initial`, `medial`, `final`

Content: The complete word containing the prolongation.

```
<!-- "Ssssssee" - prolonged initial /s/ -->
<saml:prolongation sound="s" severity="2" position="initial">
```

```
    see
</saml:prolongation>

<!-- "Misssssstake" - prolonged medial /s/ -->
<saml:prolongation sound="s" severity="2" position="medial">
    mistake
</saml:prolongation>

<!-- "Shhhheep" - prolonged digraph -->
<saml:prolongation sound="sh" severity="1" position="initial">
    sheep
</saml:prolongation>
```

Note

Prolongations can only occur on **continuant sounds**:

- Fricatives: /f/, /v/, /s/, /z/, /sh/, /zh/, /th/
- Nasals: /m/, /n/, /ng/
- Approximants: /l/, /r/, /w/, /y/
- Vowels: all vowel sounds

Warning

Note on Vowel Prolongations: Vowel prolongations are less commonly identified as core stuttering behaviors compared to consonant prolongations. Extended vowel sounds may represent:

- Normal prosodic variation or emphasis (e.g., "sooo good!")
- Thinking time or hesitation (not stuttering)
- Actual stuttering-related prolongation

Annotators should use contextual judgment when tagging vowel prolongations. Consider whether the prolongation appears involuntary and is accompanied by visible tension or struggle, which would indicate a stuttering-related dysfluency rather than normal speech variation.

5.1.4 The `<saml:block>` Element

Marks moments when speech stops or gets stuck.

Table 17: `<saml:block>` Attributes

Attribute	Type	Required	Description
`type`	Enumeration	Yes	`audible` or `inaudible`
`sound`	Enumeration	Yes	`consonant` or `vowel`
`severity`	Integer 0–8	Yes	Duration-based severity rating
`duration`	Integer	No	Actual duration in milliseconds

Content: Empty element (self-closing).

```
<!-- Inaudible block before consonant (was: IBC) -->
I want a <saml:block type="inaudible" sound="consonant"
                     severity="2"/> baby.

<!-- Audible block before consonant (was: ABC) -->
Let's <saml:block type="audible" sound="consonant"
                  severity="1"/> go.

<!-- Inaudible block before vowel (was: IBV) -->
I ate an <saml:block type="inaudible" sound="vowel"
                     severity="1"/> apple.
```

```
<!-- Audible block before vowel (was: ABV) -->
<saml:block type="audible" sound="vowel"
            severity="2"/> Every day.
```

Table 18: Block Type Mapping from Legacy Format

Legacy Tag	SAML type	SAML sound
[IBC n]	inaudible	consonant
[ABC n]	audible	consonant
[IBV n]	inaudible	vowel
[ABV n]	audible	vowel

5.2 Auxiliary Elements

5.2.1 The <saml:unintelligible> Element

Marks speech that cannot be clearly understood.

Table 19: <saml:unintelligible> Attributes

Attribute	Type	Required	Description
type	Enumeration	Yes	guess, count, unknown
value	String	Conditional	The guessed word (when type="guess")
wordCount	Integer	Conditional	Number of words (when type="count")

```
<!-- Guess at word (was: {g: word}) -->
She was my <saml:unintelligible type="guess" value="idol"/>.

<!-- Word count known (was: {w: n}) -->
It was <saml:unintelligible type="count" wordCount="3"/> yesterday.

<!-- Completely unintelligible (was: {u: }) -->
Replace <saml:unintelligible type="unknown"/> very tightly.
```

5.2.2 The <saml:overlap> Element

Marks where another speaker's speech overlaps with the main speaker.

Table 20: <saml:overlap> Attributes

Attribute	Type	Required	Description
speaker	String	No	Identifier of the overlapping speaker

```
<!-- Simple overlap marker -->
Let's go <saml:overlap/> right now. Are you coming?

<!-- With speaker identification -->
Let's go <saml:overlap speaker="speaker2"/> right now.
```

5.2.3 The <saml:letter> Element

Marks individually spoken letters (spelling out words or acronyms).

```
<!-- Spelling out a name -->
My name is spelled
```

```
<saml:letter>J</saml:letter>
<saml:letter>E</saml:letter>
<saml:letter>S</saml:letter>
<saml:letter>S</saml:letter>
<saml:letter>Y</saml:letter>
<saml:letter>E</saml:letter>.

<!-- Acronym spoken as letters -->
I watch <saml:letter>T</saml:letter><saml:letter>V</saml:letter>
every night.
```

5.2.4 The <saml:nonspeech> Element

Marks non-speech sounds produced by the speaker.

Table 21: `<saml:nonspeech>` Attributes

Attribute	Type	Required	Description
`type`	Enumeration	Yes	See list below
`description`	String	No	Additional description (for type="other")

Valid `type` values: `laugh`, `cough`, `sigh`, `cry`, `clear-throat`, `sniff`, `inhale`, `exhale`, `other`.

```
I just <saml:nonspeech type="cry"/> wanted to tell her
<saml:nonspeech type="inhale"/> that I loved her.

<saml:nonspeech type="clear-throat"/> Good morning everyone.
```

5.3 Metadata Element

5.3.1 The <saml:metadata> Element

Contains file and speaker information. Should appear at the beginning of the document.

```
<speak version="1.1"
       xmlns="http://www.w3.org/2001/10/synthesis"
       xmlns:saml="https://shunyalabs.ai/2026/saml"
       xml:lang="en-US">

    <saml:metadata>
        <saml:file id="CONV_2024_001"
                   duration="12300"
                   task="conversation"/>
        <saml:speaker id="SPK_042"
                      age-group="adult"
                      severity="moderate"/>
    </saml:metadata>

    <!-- Transcription content follows -->

</speak>
```

5.4 Severity Rating Scale

All dysfluency elements include a `severity` attribute rated 0–8 based on duration:

Table 22: Severity Rating Scale

Rating	Duration Range	Description
0	< 0.5 seconds	Very brief
1	≥ 0.5 to < 1.0 seconds	Brief
2	≥ 1.0 to < 2.0 seconds	Short
3	≥ 2.0 to < 3.0 seconds	Moderate
4	≥ 3.0 to < 5.0 seconds	Moderate-long
5	≥ 5.0 to < 10.0 seconds	Long
6	≥ 10.0 to < 30.0 seconds	Very long
7	≥ 30.0 to < 60.0 seconds	Extended
8	≥ 60.0 seconds	Extremely extended

Note

Boundary Assignment: Boundary values are assigned to the higher severity rating. For example, a dysfluency measured at exactly 1.0 seconds receives a severity rating of 2, not 1.

6 Dysfluency Annotation Guidelines

This section provides detailed rules for annotating each type of dysfluency.

Note

Quick Reference: Before diving into the classification details below, you may find it helpful to consult Table 25 (Tagging Decision Summary) in Section 8, which provides a complete summary of what to tag and what not to tag. Many annotators prefer to answer "should I tag this?" before asking "what type is it?"

6.1 Annotating Repetitions

6.1.1 Basic Rules

1. Use graphemic (written) representation from the target word
2. Include each repetition as a separate `<saml:unit>` element
3. Place the target word as text immediately after the closing tag
4. Set the `type` attribute based on what is repeated

6.1.2 Choosing the Repetition Type

Table 23: Repetition Type Selection

Type	Abbreviation	Use When / Example
`incomplete-syllable`	ISR	Any repetition that does not complete the full syllable or word. Includes single sounds and partial syllables. Examples: "d-d-d-dog", "do-do-do-dog", "d-d-do-do-dog"
`syllable`	SR	Complete syllables or monosyllabic words are repeated. Examples: "dog-dog-dog", "to-to-to-toffee", "ba-ba-baby"
`mixed`	N/A	Combination of ISR + SR **only**. Example: "t-t-to-to-toffee" (incomplete syllables followed by complete syllables)

Warning

Mixed Repetition Clarification: A repetition is classified as "mixed" **only** when there is a combination

of an Incomplete Syllable Repetition (ISR) and a Syllable Repetition (SR). When different lengths of incomplete syllables are repeated (e.g., "d-d-do-do-dog"), this should be marked as ISR, **not** as mixed repetition, because all units are still incomplete syllables.

6.1.3 Repetition Type Decision Flowchart

Use the following flowchart to quickly determine the correct repetition type when annotating:

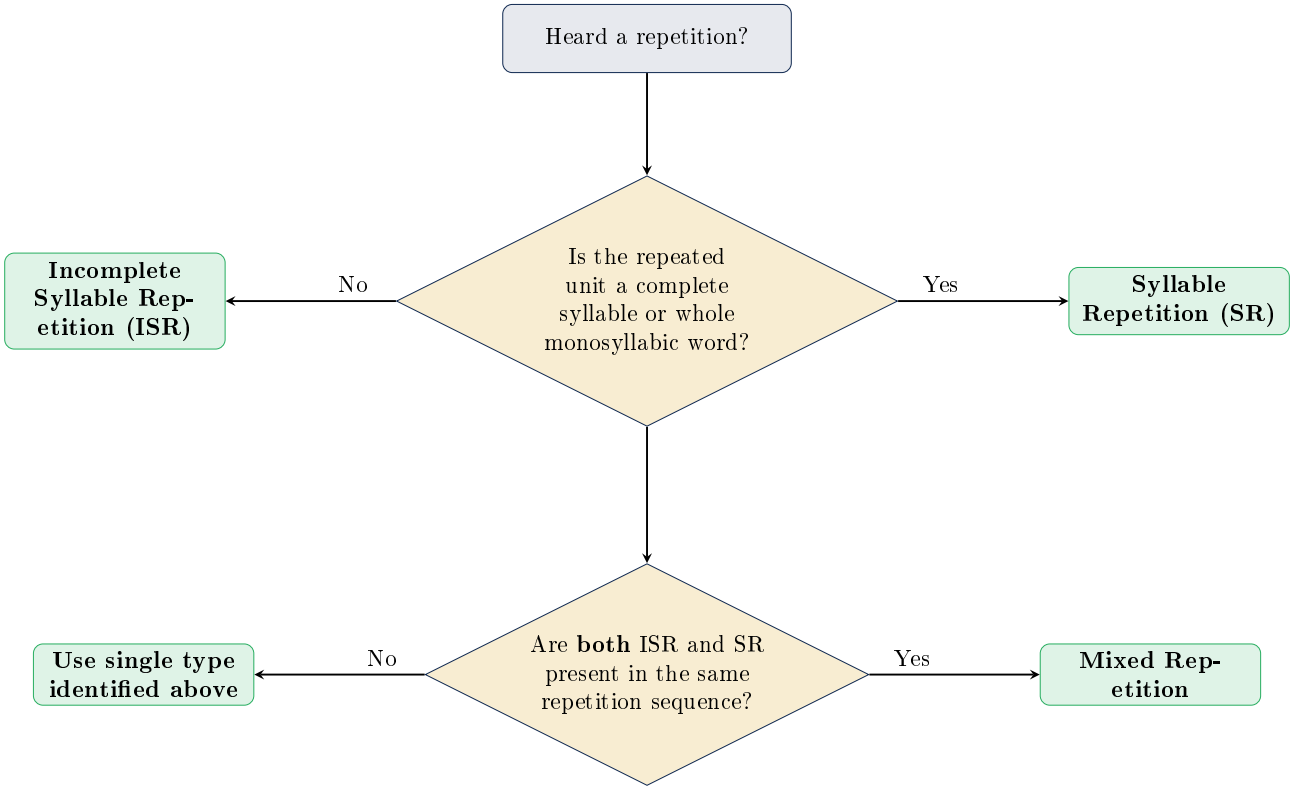

Figure 2: Repetition Type Decision Flowchart

Using the Flowchart

Example 1: "b-b-baby"

1. Is "b" a complete syllable? **No** (it's just a consonant)
2. → ISR
3. Are both ISR and SR present? **No**
4. **Result: Incomplete Syllable Repetition (ISR)**

Example 2: "ba-ba-baby"

1. Is "ba" a complete syllable of "baby"? **Yes** ("ba" is the first syllable)
2. → SR
3. Are both ISR and SR present? **No**
4. **Result: Syllable Repetition (SR)**

Example 3: "t-t-to-to-toffee"

1. Is "t" a complete syllable? **No** → ISR present
2. Is "to" a complete syllable of "toffee"? **Yes** → SR present
3. Are both ISR and SR present? **Yes**
4. **Result: Mixed Repetition**

6.1.4 Incomplete Syllable Repetitions (ISR)

When any portion less than a complete syllable is repeated (this includes single sounds/phonemes as well as partial syllables):

✓ **Correct**

```
<!-- Single sound: "c-c-car" -->
<saml:repetition type="incomplete-syllable" severity="1">
    <saml:unit>c</saml:unit>
    <saml:unit>c</saml:unit>
</saml:repetition> car

<!-- Digraph: "sh-sh-she" -->
<saml:repetition type="incomplete-syllable" severity="1">
    <saml:unit>sh</saml:unit>
    <saml:unit>sh</saml:unit>
</saml:repetition> she

<!-- Partial syllable: "do-do-dog" where "do" is not a complete syllable of
    ↪ "dog" -->
<saml:repetition type="incomplete-syllable" severity="1">
    <saml:unit>do</saml:unit>
    <saml:unit>do</saml:unit>
</saml:repetition> dog

<!-- Partial syllable: "stra-stra-strange" where "stra" is not a complete
    ↪ syllable -->
<saml:repetition type="incomplete-syllable" severity="2">
    <saml:unit>stra</saml:unit>
    <saml:unit>stra</saml:unit>
</saml:repetition> strange

<!-- Varying lengths (still ISR, NOT mixed): "d-d-do-do-dog" -->
<saml:repetition type="incomplete-syllable" severity="2">
    <saml:unit>d</saml:unit>
    <saml:unit>d</saml:unit>
    <saml:unit>do</saml:unit>
    <saml:unit>do</saml:unit>
</saml:repetition> dog
```

Note

ISR includes all of the following:

- Single phoneme repetitions (e.g., “d-d-dog”)
- Partial syllable repetitions (e.g., “do-do-dog” where “dog” is monosyllabic)
- Combinations of different incomplete syllable lengths (e.g., “d-d-do-do-dog”)

The key criterion is that the repeated unit does **not** form a complete syllable of the target word.

Warning

Consonant Clusters and Syllabification Ambiguity

When annotating repetitions involving consonant clusters or words with ambiguous syllable boundaries, follow these guidelines:

Consonant Clusters:

- Single consonant repeated: `type="incomplete-syllable"` (e.g., “s-s-strange”)
- Partial cluster repeated: `type="incomplete-syllable"` (e.g., “str-str-strange”)
- Full cluster as onset repeated: `type="incomplete-syllable"` (e.g., “stra-stra-strange”)

Syllabification Reference: For consistent syllable boundary determination, annotators should use the Carnegie Mellon Pronouncing Dictionary (CMUdict) or equivalent pronunciation dictionary for their target language. When the dictionary disagrees with perceived syllabification in the audio, defer to the acoustic realization.

Spoken Pattern	Type	Rationale
"s-s-strong"	incomplete-syllable	Single phoneme /s/ (ISR)
"str-str-strong"	incomplete-syllable	Partial onset cluster (ISR)
"pro-pro-problem"	syllable	Complete first syllable (SR)
"prob-prob-problem"	incomplete-syllable	Exceeds syllable boundary (ISR)

× **Incorrect**

```
<!-- WRONG: Using phonetic spelling instead of grapheme -->
<saml:repetition type="incomplete-syllable" severity="1">
    <saml:unit>k</saml:unit>
    <saml:unit>k</saml:unit>
</saml:repetition> car

<!-- WRONG: Incomplete digraph -->
<saml:repetition type="incomplete-syllable" severity="1">
    <saml:unit>s</saml:unit>
    <saml:unit>s</saml:unit>
</saml:repetition> she
```

6.1.5 Syllable Repetitions (SR)

When complete syllables or monosyllabic words are repeated:

✓ **Correct**

```
<!-- Complete syllable: "ba-ba-ba-baby" -->
<saml:repetition type="syllable" severity="2" count="4">
    <saml:unit>ba</saml:unit>
    <saml:unit>ba</saml:unit>
    <saml:unit>ba</saml:unit>
</saml:repetition> baby

<!-- Complete syllable: "ci-ci-cigarette" -->
<saml:repetition type="syllable" severity="1">
    <saml:unit>ci</saml:unit>
    <saml:unit>ci</saml:unit>
</saml:repetition> cigarette

<!-- Monosyllabic word: "dog-dog-dog" -->
<saml:repetition type="syllable" severity="1">
    <saml:unit>dog</saml:unit>
    <saml:unit>dog</saml:unit>
</saml:repetition> dog

<!-- Monosyllabic word: "to-to-to-toffee" -->
<saml:repetition type="syllable" severity="2">
    <saml:unit>to</saml:unit>
    <saml:unit>to</saml:unit>
    <saml:unit>to</saml:unit>
</saml:repetition> toffee
```

× **Incorrect**

```
<!-- WRONG: Using phonetic spelling -->
<saml:repetition type="syllable" severity="1">
```

```
    <saml:unit>si</saml:unit>
    <saml:unit>si</saml:unit>
</saml:repetition> cigarette
```

Note

Monosyllabic Word Repetitions: Repetitions of monosyllabic words (e.g., "I-I-I", "the-the-the", "dog-dog-dog") are now classified as **Syllable Repetitions (SR)** because the entire word constitutes a single syllable.

Warning

Do NOT tag multisyllabic word repetitions. These are normal disfluencies:

```
<!-- NOT tagged - normal disfluency -->
because because I was late

<!-- NOT tagged - normal disfluency -->
maybe maybe we should go
```

6.1.6 Mixed Repetitions

When **both** incomplete syllables (ISR) **and** complete syllables (SR) are repeated in sequence:

✓ **Correct**

```
<!-- "t-t-to-to-toffee" - ISR (t-t) followed by SR (to-to) -->
<saml:repetition type="mixed" severity="2">
    <saml:unit>t</saml:unit>
    <saml:unit>t</saml:unit>
    <saml:unit>to</saml:unit>
    <saml:unit>to</saml:unit>
</saml:repetition> toffee
```

Warning

Important Distinction: Do **not** classify varying lengths of incomplete syllables as "mixed." For example, "d-d-do-do-dog" should be marked as `type="incomplete-syllable"` because all repeated units are incomplete syllables (none are complete syllables). Mixed repetition requires a transition from incomplete syllables to complete syllables.

6.2 Annotating Prolongations

6.2.1 Basic Rules

1. Only mark prolongations on continuant sounds
2. Set `sound` to the prolonged grapheme(s)
3. Include the complete target word as element content
4. Use `position` to indicate where in the word the prolongation occurs

6.2.2 Word-Initial Prolongations

✓ **Correct**

```
<!-- "Ssssssee" -->
<saml:prolongation sound="s" severity="2" position="initial">
```

```
    see
</saml:prolongation>

<!-- "Mmmmom" -->
<saml:prolongation sound="m" severity="1" position="initial">
    mom
</saml:prolongation>
```

6.2.3 Word-Medial Prolongations

✓ **Correct**

```
<!-- "Misssssstake" -->
<saml:prolongation sound="s" severity="2" position="medial">
    mistake
</saml:prolongation>

<!-- "Ammmazing" -->
<saml:prolongation sound="m" severity="1" position="medial">
    amazing
</saml:prolongation>
```

6.2.4 Digraph Prolongations

When the prolonged sound is written with two letters:

✓ **Correct**

```
<!-- "Shhhheep" -->
<saml:prolongation sound="sh" severity="1" position="initial">
    sheep
</saml:prolongation>

<!-- "Thhhhat" -->
<saml:prolongation sound="th" severity="1" position="initial">
    that
</saml:prolongation>
```

× **Incorrect**

```
<!-- WRONG: Incomplete digraph -->
<saml:prolongation sound="s" severity="1" position="initial">
    sheep
</saml:prolongation>
```

6.2.5 Word-Final Prolongations

While less common, prolongations can occur at the end of words:

✓ **Correct**

```
<!-- "Yesssss" - prolonged final /s/ -->
<saml:prolongation sound="s" severity="2" position="final">
    yes
</saml:prolongation>

```

```
<!-- "Goooo" - prolonged final vowel -->
<saml:prolongation sound="o" severity="1" position="final">
    go
</saml:prolongation>
```

Note

Word-final prolongations may sometimes reflect emphasis rather than stuttering. Apply contextual judgment as described in Section 8.

6.3 Annotating Blocks

6.3.1 Identifying Blocks

A block occurs when:

- Speech stops completely (inaudible)
- You hear effort or tension but no speech sound (audible)
- The stoppage is clearly involuntary

6.3.2 Choosing Block Attributes

To determine the correct block type, answer two questions:

1. **Question 1**: Can you hear sound or tension during the block?
   - **No** (complete silence) → *Inaudible*
   - **Yes** (audible struggle, tension, or partial sound) → *Audible*
2. **Question 2**: What is the next sound after the block?
   - Starts with a consonant sound (b, d, g, p, t, k, etc.) → *Consonant*
   - Starts with a vowel sound (a, e, i, o, u) → *Vowel*

Warning

Silent Letter Exception: Block classification must be based on the **phonetic onset** (the actual sound) of the word, **not** its orthographic spelling. Words with silent letters at the beginning should be classified based on the sound that is actually produced:

- "honest" → begins with vowel sound /o/, classify as *Vowel* (not consonant despite "h")
- "hour" → begins with vowel sound /au/, classify as *Vowel*
- "heir" → begins with vowel sound /e/, classify as *Vowel*
- "knight" → begins with consonant sound /n/, classify as *Consonant* (not "k")
- "write" → begins with consonant sound /r/, classify as *Consonant* (not "w")
- "gnome" → begins with consonant sound /n/, classify as *Consonant* (not "g")

Table 24: Block Type Selection Matrix

	Next sound: Consonant	**Next sound: Vowel**
Inaudible (silent)	IBC	IBV
Audible (hear tension)	ABC	ABV

6.3.3 Block Type Decision Flowchart

Use this flowchart to quickly determine the correct block type:

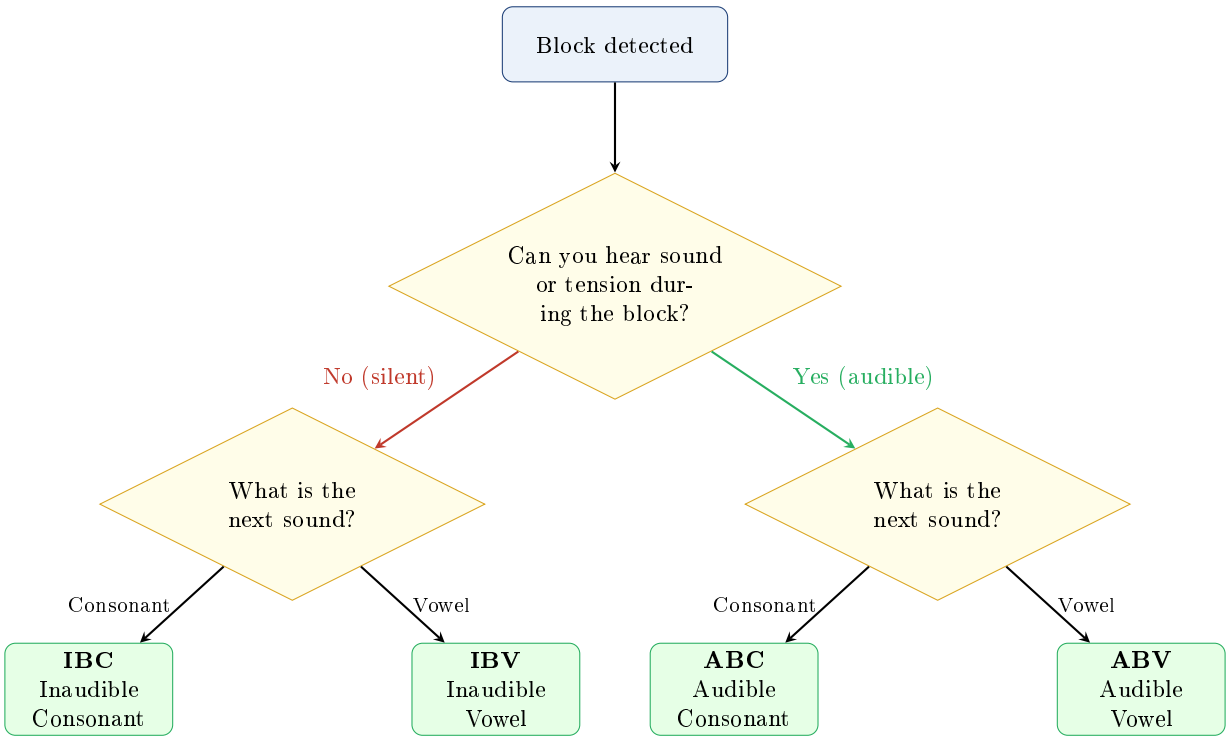

Figure 3: Block Type Decision Flowchart

Using the Block Flowchart

What You Observe	Flowchart Path	Result
Silent pause before "baby"	No → Consonant	IBC
Tense struggle before "go"	Yes → Consonant	ABC
Silent pause before "apple"	No → Vowel	IBV
Audible glottal tension before "every"	Yes → Vowel	ABV
Silent before "honest"	No → Vowel (phonetic /o/)	IBV

Block Type Examples

Scenario	Description	Block Type
"...a [silence] baby"	Silent block before /b/	IBC
"...to [struggle] go"	Audible tension before /g/	ABC
"...an [silence] apple"	Silent block before /a/	IBV
"[tension] Every day"	Audible tension before /e/	ABV

6.3.4 Block Examples

Inaudible Block on Consonant

Context: Complete silence before the /b/ in "baby"

```
I want a <saml:block type="inaudible" sound="consonant"
                     severity="2"/> baby.
```

Audible Block on Consonant

Context: Audible tension before the /g/ in "go"

```
Let's <saml:block type="audible" sound="consonant"
                  severity="1"/> go.
```

Inaudible Block on Vowel

Context: Complete silence before the /a/ in "apple"

```
I ate an <saml:block type="inaudible" sound="vowel"
                     severity="1"/> apple.
```

Audible Block on Vowel

Context: Audible glottal tension before "every"

```
<saml:block type="audible" sound="vowel"
            severity="2"/> Every day.
```

6.4 Consecutive Dysfluencies

Multiple dysfluencies can occur together. Tag each one separately:

Block followed by Repetition

```
<!-- Block before repetition on same word -->
I want a <saml:block type="inaudible" sound="consonant"
                     severity="1"/>
<saml:repetition type="syllable" severity="1">
    <saml:unit>ba</saml:unit>
    <saml:unit>ba</saml:unit>
</saml:repetition> baby.
```

Self-Correction During Dysfluency

When a speaker abandons a word during a dysfluency and restarts with a different word, annotate the dysfluency on the abandoned word and transcribe the correction as plain text:

```
<!-- Speaker starts "baby" but switches to "child" -->
I want a <saml:repetition type="syllable" severity="1">
    <saml:unit>ba</saml:unit>
    <saml:unit>ba</saml:unit>
</saml:repetition> [...] child.
```

Note: The ellipsis in brackets `[...]` indicates the abandoned word. The target word "baby" is not written because the speaker never completed it. If the intended word is unclear, use the unintelligible element:

```
<!-- Unclear intended word with self-correction -->
I want a <saml:repetition type="syllable" severity="1">
    <saml:unit>ba</saml:unit>
    <saml:unit>ba</saml:unit>
```

```
</saml:repetition> <saml:unintelligible type="guess" value="baby"/> [...]
    ↪ child.
```

Prolongation with Subsequent Block

```
<!-- "Sssss-[block]-see" - prolongation followed by block -->
<saml:prolongation sound="s" severity="2" position="initial">
    see
</saml:prolongation>
<saml:block type="inaudible" sound="vowel" severity="1"/> see
```

6.5 Dysfluency with Unintelligible Speech

```
<!-- Repetition followed by guessed word -->
<saml:repetition type="incomplete-syllable" severity="1">
    <saml:unit>a</saml:unit>
    <saml:unit>a</saml:unit>
</saml:repetition> <saml:unintelligible type="guess" value="available"/>

<!-- Multiple unintelligible elements -->
I am <saml:repetition type="incomplete-syllable" severity="1">
    <saml:unit>a</saml:unit>
    <saml:unit>a</saml:unit>
</saml:repetition> <saml:unintelligible type="guess" value="available"/>
for this <saml:unintelligible type="unknown"/>
```

7 Severity Measurement Protocol

7.1 General Principles

1. Use audio software with waveform display (Audacity, Praat)
2. Measure at millisecond precision when possible
3. Round to one decimal place before applying severity scale
4. When in doubt, use the lower severity rating
5. Optionally record exact duration in the `duration` attribute (in milliseconds)

7.2 Measurement Points

7.2.1 For Repetitions

- **Start**: Beginning of the first repetition unit
- **End**: End of the last repetition unit (before target word)

7.2.2 For Prolongations

- **Start**: The beginning point of the sound itself (not when it "becomes abnormal")
- **End**: The ending point of the prolonged sound, i.e., the point where the sound stops being heard

Note

Rationale for Objective Measurement Points:

- **Start point**: Determining when a sound "begins to extend beyond normal" is subjective and cannot be reliably identified either perceptually or via waveform/spectrogram analysis. Using the beginning of the sound itself provides an objective, consistent measurement.

- **End point**: Using "when the speaker moves to the next sound" can incorrectly include periods of silence (e.g., in cases where a prolongation is followed by silence before a fluent word). The end point should be where the prolonged sound itself stops being heard.

Prolongation Measurement Edge Case

Scenario: Speaker produces "sssss" (prolonged /s/), followed by 500ms of silence, then says "see" fluently.

Correct measurement: Start at beginning of /s/ sound, end when /s/ sound stops (before the silence).

Incorrect measurement: Including the 500ms silence in the prolongation duration would falsely inflate the severity rating.

7.2.3 For Blocks

- **Start**: When speech stops (after last fluent sound)
- **End**: When fluent speech resumes

7.3 Recording Exact Duration

While `severity` is required, you can also record exact duration:

```
<!-- Severity 2 with exact duration of 1500ms -->
<saml:repetition type="syllable" severity="2" duration="1500">
    <saml:unit>ba</saml:unit>
    <saml:unit>ba</saml:unit>
</saml:repetition> baby
```

8 Distinguishing Stuttering from Typical Disfluencies

8.1 What NOT to Tag

The following are normal disfluencies and should **not** be tagged:

8.1.1 Interjections and Filler Words

Write as plain text:

```
<!-- Plain text, no tags -->
Um, I think so.
It was, uh, good.
```

8.1.2 Phrase Repetitions

```
<!-- Plain text, no tags -->
I want to I want to go.
```

8.1.3 Multisyllabic Word Repetitions

```
<!-- Plain text, no tags -->
Because because I was tired.
```

8.1.4 False Starts and Revisions

```
<!-- Plain text, no tags -->
I went to the... I drove to the store.
```

8.1.5 Intentional Prolongations

```
<!-- Plain text - this is emphasis, not stuttering -->
It was sooo good!
```

8.1.6 Intentional Repetitions

Repetitions that are purposeful (e.g., for emphasis, while thinking, or self-correction) should **not** be tagged as dysfluencies:

```
<!-- Plain text - thinking/hesitation, not stuttering -->
I, I was thinking to go to the mall.

<!-- Plain text - emphasis, not stuttering -->
No, no, no, that's not what I meant.

<!-- Plain text - self-correction, not stuttering -->
I went to the, the store, I mean the mall.
```

Note

Distinguishing Intentional from Stuttering Repetitions: Consider the context and manner of the repetition:

- **Intentional**: Occurs at natural pause points, no visible tension, speaker appears in control, often accompanied by thinking or self-correction
- **Stuttering**: Occurs involuntarily, may show visible tension or struggle, speaker appears to lose control momentarily, often on specific sounds rather than whole words

When in doubt, consider whether the repetition appears effortful or involuntary.

8.2 Decision Summary

Table 25: Tagging Decision Summary

Dysfluency Type	Tag?	Element / Type
Incomplete syllable repetition (b-b-baby, do-do-dog)	Yes	`<saml:repetition>` type="incomplete-syllable"
Syllable repetition (ba-ba-baby)	Yes	`<saml:repetition>` type="syllable"
Monosyllabic word repetition (I-I-I, dog-dog-dog)	Yes	`<saml:repetition>` type="syllable"
Mixed repetition (t-t-to-to-toffee)	Yes	`<saml:repetition>` type="mixed"
Multisyllabic word repetition	No	N/A
Phrase repetition	No	N/A
Intentional repetitions (I, I was thinking)	No	N/A
Involuntary prolongation	Yes	`<saml:prolongation>`
Intentional emphasis prolongation	No	N/A
Block (speech stops)	Yes	`<saml:block>`
Natural pause	No	N/A
Interjection (um, uh)	No	N/A
Revision/false start	No	N/A

9

Transcription Conventions

9.1 Basic Principles

1. Capture everything exactly as spoken, including false starts, filler words, and repetitions
2. Use standard American English spelling, capitalization, and punctuation
3. Never correct grammar or paraphrase; transcribe exactly what you hear
4. Do not break long sentences unless the speaker clearly pauses for a new thought
5. Use appropriate XML escaping for special characters

9.2 XML Special Characters

Table 26: XML Character Escaping

Character	Escape Sequence	Example
`<`	`<`	"less than"
`>`	`>`	"greater than"
`&`	`&`	"Tom & Jerry"
`"`	`"`	Inside attributes
`'`	`'`	Inside attributes

9.3 Numbers

Write numbers as words:

```
I have twenty-three dollars.
She is forty-two years old.
```

9.4 Filler Words

Standard spellings (single final letter):

Table 27: Filler Word Spellings (Quick Reference)

Sound	Spelling
Hesitation with schwa	uh
Hesitation with /m/	um
Realization	ah, oh
Agreement	mm-hmm, uh-huh
Disagreement	mm-mm, uh-uh
Questioning	eh, huh
Thinking	hmm
Casual affirmative/negative	yeah, nah
Other	er, erm, ew

Note

For a comprehensive list of interjections, discourse markers, and agreement sounds with usage examples, see the Detailed Interjection Reference in Appendix E.

9.5 Non-Speech Sounds

Use the `<saml:nonspeech>` element for sounds produced by the speaker that are not speech:

```
<saml:nonspeech type="clear-throat"/> Good morning.
I just <saml:nonspeech type="laugh"/> can't believe it.
```

Standard types: `laugh`, `cough`, `sigh`, `cry`, `clear-throat`, `sniff`, `inhale`, `exhale`, `other`.

Note

For a comprehensive list of non-speech sounds, usage examples, and the distinction between speaker-produced sounds (annotate) vs. background sounds (do not annotate), see the Detailed Non-Speech Sound Reference in Appendix E.

10 Special Situations

10.1 Multiple Speakers

Use separate <p> elements with speaker identification:

```
<speak version="1.1"
       xmlns="http://www.w3.org/2001/10/synthesis"
       xmlns:saml="https://shunyalabs.ai/2026/saml"
       xml:lang="en-US">

    <p id="speaker1">Hello, how are you?</p>

    <p id="speaker2">I'm <saml:repetition type="incomplete-syllable"
        ↪ severity="1">
        <saml:unit>d</saml:unit>
        <saml:unit>d</saml:unit>
    </saml:repetition> doing well, thank you.</p>

    <p id="speaker1">That's great to hear.</p>

</speak>
```

10.2 Overlapping Speech

```
<p id="speaker1">Let's go <saml:overlap speaker="speaker2"/>
    right now. Are you coming with me?</p>
```

10.3 IVR and Automated Speech

Transcribe as a separate speaker without dysfluency tags:

```
<p id="ivr">Please say or enter your account number.</p>

<p id="speaker1">My account number is
    <saml:repetition type="incomplete-syllable" severity="1">
        <saml:unit>s</saml:unit>
        <saml:unit>s</saml:unit>
    </saml:repetition> seven two...</p>
```

10.4 Combining with Standard SSML

SAML elements work alongside standard SSML elements:

```
<speak version="1.1"
       xmlns="http://www.w3.org/2001/10/synthesis"
       xmlns:saml="https://shunyalabs.ai/2026/saml"
       xml:lang="en-US">

    <!-- Using SSML prosody with SAML dysfluency -->
    <prosody rate="slow">
```

```
        I <saml:block type="audible" sound="consonant"
                      severity="1"/> think we should
        <emphasis level="strong">really</emphasis> consider this.
    </prosody>

    <!-- Using SSML break for intentional pause -->
    Thank you for listening. <break time="500ms"/>
    Any questions?

</speak>
```

10.5 Background Sound Annotation

Note

Decision: Background sounds SHOULD be annotated.

Background sounds provide important contextual information for ASR model training and can help explain speech patterns, unintelligible segments, or speaker behavior. Annotating background sounds helps:

- **ASR robustness**: Models can learn to handle speech in noisy environments
- **Context understanding**: Explains why certain speech segments may be affected
- **Data quality**: Allows filtering or weighting of training data by acoustic conditions
- **Clinical relevance**: Environmental factors may influence stuttering severity

10.5.1 How to Annotate Background Sounds

Use the `<saml:background>` element to mark background sounds. This element can appear at the beginning of an utterance or inline where the background sound is notable.

```
<!-- Background sound at start of utterance -->
<saml:background type="traffic" level="moderate"/>
I was walking to the store.

<!-- Background sound appearing mid-utterance -->
I was walking <saml:background type="siren" level="loud"/>
to the store when the ambulance passed.

<!-- Multiple background sounds -->
<saml:background type="music" level="soft"/>
<saml:background type="crowd" level="moderate"/>
So I was at the party and...
```

10.5.2 Background Sound Attributes

Table 28: `<saml:background>` Attributes

Attribute	Type	Required	Description
`type`	String	Yes	Category of background sound (see list below)
`level`	Enumeration	No	`soft`, `moderate`, `loud`, `overwhelming`
`description`	String	No	Additional description if type="other"
`continuous`	Boolean	No	Whether sound persists throughout (default: true)

10.5.3 Background Sound Types

Table 29: Background Sound Type Values

Type Value	Category	Examples
Traffic & Transportation		
`traffic`	Vehicles	General road noise, cars passing
`siren`	Emergency	Ambulance, police, fire truck
`horn`	Vehicles	Car horns, truck horns
`aircraft`	Aviation	Airplane overhead, helicopter
`train`	Rail	Train passing, subway
Environmental		
`wind`	Weather	Wind noise, gusts
`rain`	Weather	Rain, storm sounds
`thunder`	Weather	Thunder, lightning
`birds`	Nature	Bird calls, chirping
`animals`	Nature	Dogs barking, other animals
`water`	Nature	Running water, waves, fountain
Human Activity		
`crowd`	People	Multiple people talking, crowd noise
`conversation`	People	Distinct other conversation nearby
`children`	People	Children playing, crying
`footsteps`	Movement	Walking, running sounds
Indoor/Household		
`hvac`	Climate	Air conditioning, heating, fans
`appliance`	Household	Refrigerator, washer, microwave
`tv`	Media	Television audio in background
`music`	Media	Background music, radio
`phone`	Communication	Phone ringing, notification sounds
`door`	Structure	Doors opening/closing, knocking
Technical/Equipment		
`static`	Recording	Audio static, interference
`hum`	Electrical	Electrical hum, buzz
`echo`	Acoustic	Room echo, reverberation
`mic-noise`	Recording	Microphone handling, wind on mic
`other`	Miscellaneous	Use with `description` attribute

10.5.4 Background Sound Level Guide

Table 30: Background Sound Level Descriptions

Level	Description
`soft`	Barely noticeable; does not affect speech comprehension
`moderate`	Clearly audible but speech remains fully intelligible
`loud`	Competes with speech; some words may be harder to hear
`overwhelming`	Dominates the audio; speech significantly obscured

10.5.5 Shorthand Notation for Background Sounds

Shorthand	SAML Element
[bg:traffic]	<saml:background type="traffic"/>
[bg:traffic loud]	<saml:background type="traffic" level="loud"/>
[bg:music soft]	<saml:background type="music" level="soft"/>
[bg:crowd moderate]	<saml:background type="crowd" level="moderate"/>
[bg:other “dog barking”]	<saml:background type="other" description="dog barking"/>

10.5.6 When to Annotate Background Sounds

- **Always annotate** when background sound is at “moderate” level or higher
- **Annotate at start** of chunk/utterance if sound is continuous throughout
- **Annotate inline** if sound appears suddenly or changes
- **Note level changes** if background sound intensity changes significantly
- **Use with unintelligible** if background sound causes speech to be unclear

Background Sound with Unintelligible Speech

```
<saml:background type="siren" level="loud"/>
I was trying to <saml:unintelligible type="unknown"/>
when the ambulance went by.
```

Note

Distinction from Non-Speech Sounds: Background sounds are environmental/external sounds **not produced by the speaker**. Speaker-produced sounds (coughs, laughs, sighs) use `<saml:nonspeech>`. If the speaker coughs, use `<saml:nonspeech type="cough"/>`. If someone else coughs nearby, you may note it as `<saml:background type="other" description="cough nearby"/>`.

11 Shorthand Notation for Annotators

For faster manual annotation, SAML defines a shorthand notation that can be converted to full XML.

11.1 Shorthand Syntax

Table 31: Shorthand to XML Mapping

Shorthand	Full XML
`( [R:isyl 1] b- b- ) baby`	`<saml:repetition>` with type="incomplete-syllable"
`( [R:isyl 1] do- do- ) dog`	`<saml:repetition>` with type="incomplete-syllable"
`( [R:syl 1] ba- ba- ) baby`	`<saml:repetition>` with type="syllable"
`( [R:syl 1] dog- dog- ) dog`	`<saml:repetition>` with type="syllable"
`( [R:mix 2] t- t- to- to- ) toffee`	`<saml:repetition>` with type="mixed"
`( [P 2] s- )ee`	`<saml:prolongation>` with sound="s"
`( [IBC 2] ) baby`	`<saml:block>` type="inaudible" sound="consonant"
`( [ABC 1] ) go`	`<saml:block>` type="audible" sound="consonant"
`( [IBV 1] ) apple`	`<saml:block>` type="inaudible" sound="vowel"
`( [ABV 2] ) every`	`<saml:block>` type="audible" sound="vowel"
`{g: word}`	`<saml:unintelligible>` type="guess"
`{w: 3}`	`<saml:unintelligible>` type="count"
`{u: }`	`<saml:unintelligible>` type="unknown"
`<overlap>`	`<saml:overlap>`
`~A`	`<saml:letter>`
`[laughs]`	`<saml:nonspeech>` type="laugh"
`[coughs]`	`<saml:nonspeech>` type="cough"
`[bg:traffic]`	`<saml:background>` type="traffic"
`[bg:music soft]`	`<saml:background>` type="music" level="soft"

Note

The generic block tag `[B n]` can still be used when the annotator identifies a block but the specific type (audible/inaudible, consonant/vowel) is unclear. During conversion to XML, the converter should attempt to infer the type from context or flag it for review.

11.2 Shorthand Rules

1. Parentheses `( )` enclose dysfluency content
2. Square brackets `[ ]` contain tag type and severity
3. One space after opening parenthesis and before closing
4. Hyphens after each repeated/prolonged unit
5. Block tags have empty content
6. Repetition tags must include a type marker after the colon:
   - `:isyl` = incomplete-syllable (ISR): includes single sounds and partial syllables
   - `:syl` = syllable (SR): includes complete syllables and monosyllabic words
   - `:mix` = mixed: combination of ISR + SR only

11.3 Quick Reference: Good vs. Bad Transcription

✓ **Correct**

```
I ( [IBC 2] ) think the m( [P 1] m- )ovie was ( [R:isyl 1] re- re- ) really
    ↪ good.
```

Properly captures block, prolongation, and repetition with correct formatting and explicit type markers. Note: "re-re-" is an incomplete syllable repetition (ISR) since "re" is not a complete syllable of "really."

× **Incorrect**

```
I... think the movie was really good.
```

Disfluencies removed or replaced with punctuation. This loses critical data.

11.4 Shorthand Examples

```
# Shorthand annotation with explicit type markers
I ( [R:isyl 1] w- w- ) was going to the store, um, and I
s( [P 2] s- )aw my ( [IBC 1] ) friend.

# Converts to full SAML XML automatically
```

11.5 Side-by-Side Comparison: Shorthand and XML

The following examples show the relationship between shorthand notation and the corresponding full SAML XML output.

Incomplete Syllable Repetition (ISR): Shorthand to XML

Shorthand:

```
I ( [R:isyl 1] w- w- ) want to go.
```

Corresponding SAML XML:

```
I <saml:repetition type="incomplete-syllable" severity="1">
    <saml:unit>w</saml:unit>
    <saml:unit>w</saml:unit>
</saml:repetition> want to go.
```

Syllable Repetition: Shorthand to XML

Shorthand:

```
The ( [R:syl 2] ba- ba- ba- ) baby is sleeping.
```

Corresponding SAML XML:

```
The <saml:repetition type="syllable" severity="2">
    <saml:unit>ba</saml:unit>
    <saml:unit>ba</saml:unit>
    <saml:unit>ba</saml:unit>
</saml:repetition> baby is sleeping.
```

Prolongation: Shorthand to XML

Shorthand:

```
I ( [P 2] s- )aw the movie.
```

Corresponding SAML XML:

```
I <saml:prolongation sound="s" severity="2" position="initial">
```

```
    saw
</saml:prolongation> the movie.
```

Block: Shorthand to XML

Shorthand:

```
Let's ( [IBC 2] ) go to the store.
```

Corresponding SAML XML:

```
Let's <saml:block type="inaudible" sound="consonant"
                  severity="2"/> go to the store.
```

Complex Example: Multiple Dysfluencies

Shorthand:

```
So ( [IBC 1] ) ( [R:syl 1] I- I- ) I went to the
( [P 2] s- )tore yesterday.
```

Corresponding SAML XML:

```
So <saml:block type="inaudible" sound="consonant"
               severity="1"/>
<saml:repetition type="syllable" severity="1">
    <saml:unit>I</saml:unit>
    <saml:unit>I</saml:unit>
</saml:repetition> I went to the
<saml:prolongation sound="s" severity="2" position="initial">
    store
</saml:prolongation> yesterday.
```

Note: "I" is a monosyllabic word, so it is classified as a syllable repetition (SR).

11.6 Reference Conversion Tool

A reference implementation of the shorthand-to-XML converter is available as part of the SAML toolkit. The converter (`converter.py`) provides the following capabilities:

- Accept shorthand-annotated text files as input
- Output valid SAML/SSML XML documents with proper namespace declarations
- Validate output against the SAML XSD (requires `lxml` library)
- Support batch processing for corpus annotation
- Available as both a command-line tool and an importable Python library
- Multi-speaker transcript support with automatic paragraph generation
- Comprehensive dysfluency statistics reporting
- Strict mode for rejecting legacy syntax

11.6.1 Command-Line Usage

```
# Convert a single file
python3 converter.py input.txt -o output.xml

# Convert with metadata
python3 converter.py input.txt -o output.xml \
    --file-id recording-001 \
```

```
    --speaker-id SPK01 \
    --task conversation \
    --speaker-age adult \
    --speaker-severity moderate

# Batch convert a directory
python3 converter.py --batch input_dir/ output_dir/

# Validate output against schema
python3 converter.py input.txt -o output.xml \
    --validate --schema saml.xsd

# Show dysfluency statistics
python3 converter.py input.txt --stats
```

11.6.2 Library Usage

```
from converter import ShorthandConverter, convert_file

# Convert text directly
converter = ShorthandConverter(
    file_id="doc-001",
    speaker_id="SPK01",
    task="reading",
    speaker_severity="moderate"
)
result = converter.convert(
    "I ( [R:isyl 1] w- w- ) was going."
)
print(result.xml)

# Check for warnings/errors
if result.warnings:
    print("Warnings:", result.warnings)

# Get statistics
print(converter.get_statistics_report())
```

11.6.3 Metadata Options

The converter supports comprehensive metadata for file and speaker information:

Table 32: Converter Metadata Options

Category	Option	Values
File	`-file-id`	Unique identifier for the recording
	`-file-duration`	Duration in milliseconds
	`-task`	reading, monologue, conversation, phone, presentation, spontaneous, other
Speaker	`-speaker-id`	Unique speaker identifier
	`-speaker-age`	child, adolescent, adult
	`-speaker-severity`	mild, moderate, severe, very-severe, unassessed
	`-speaker-gender`	male, female, other, unspecified

Note

The converter is available at `https://shunyalabs.ai/saml/tools`. For detailed usage documentation and

the complete source code, see the `shorthand/` directory in the SAML repository. The converter follows the conversion rules specified in Appendix C.

12 Quality Assurance

12.1 Validation

All SAML documents should be validated against the XML Schema before submission:

```
# Example validation command
xmllint --schema saml.xsd annotation.xml --noout
```

12.2 Accuracy Requirements

Given the clinical application of this dataset, quality thresholds are set to ensure reliability for diagnostic and therapeutic use.

Table 33: Quality Thresholds

Metric	Requirement
XML validation	100% valid documents
Overall accuracy	$\geq$ 95% at batch level
Dysfluency detection	$\geq$ 90% recall, $\geq$ 90% precision
Severity rating	Exact match or within ± 1 of gold standard
Transcription accuracy	$\geq$ 98% word accuracy (WER $\leq$ 2%)
Uniqueness	Each transcript must be unique (no duplicates)
Metadata accuracy	Metadata must match the audio file exactly

12.3 Self-Review Checklist

Before submitting your annotation, verify:

- ☐ Document validates against XSD (if using full XML)
- ☐ Namespace declarations are correct
- ☐ All required attributes are present
- ☐ Severity ratings match measured durations
- ☐ Graphemic representations match target word spelling
- ☐ Correct repetition type used:
  - ISR (`incomplete-syllable`) for single sounds and partial syllables
  - SR (`syllable`) for complete syllables and monosyllabic words
  - `mixed` only for ISR + SR combinations
- ☐ Typical disfluencies (um, phrase repetitions, multisyllabic word repetitions) are NOT tagged
- ☐ Intentional repetitions (for emphasis, thinking, self-correction) are NOT tagged
- ☐ Interjections are written in running text without tags
- ☐ Hyphens are present after all repeated/prolonged units
- ☐ Correct spacing: one space after `(` and before `)`
- ☐ Speaker labels are consistent
- ☐ Special characters are properly escaped
- ☐ Metadata matches the audio file exactly
- ☐ No duplicate transcripts in the batch
- ☐ Long sentences are not artificially broken

12.4 Common Errors

Table 34: Common Annotation Errors

Error	Problem	Solution
Missing namespace	Elements not recognized	Add `xmlns:saml` declaration
Wrong attribute type	Validation fails	Check XSD for valid values
Phonetic spelling	Inconsistent data	Use graphemic spelling from word
Tagging normal disfluency	Over-annotation	Review Section 8 (disfluencies are normal)
Missing severity	Required attribute	Always include severity 0–8
Unescaped characters	XML parse error	Use `&`, `<`, etc.

12.5 Inter-Rater Reliability Protocol

To ensure annotation consistency across multiple annotators, a formal inter-rater reliability (IRR) protocol must be followed.

12.5.1 Dual Annotation Requirements

Table 35: Dual Annotation Requirements

Parameter	Requirement
Minimum dual-annotated data	20% of total corpus
Selection method	Random sampling across speakers and tasks
Annotator independence	Annotators must not discuss samples before completion
Timing	IRR checks at project start, mid-point, and end

12.5.2 Agreement Metrics

Different aspects of annotation require different agreement metrics. Given the clinical nature of this dataset, thresholds are set conservatively to ensure reliability for diagnostic and therapeutic applications.

Table 36: Inter-Rater Agreement Metrics and Thresholds

Annotation Aspect	Metric	Minimum	Target
Dysfluency presence (yes/no)	Cohen's κ	≥ 0.85	≥ 0.90
Dysfluency type (R/P/B)	Cohen's κ	≥ 0.80	≥ 0.90
Block subtype (IBC/ABC/IBV/ABV)	Cohen's κ	≥ 0.75	≥ 0.85
Severity rating (0–8)	Weighted κ	≥ 0.85	≥ 0.90
Severity rating (0–8)	ICC (two-way, absolute)	≥ 0.90	≥ 0.95
Boundary placement (ms)	Mean absolute difference	$\leq$ 50ms	$\leq$ 25ms
Transcription	Word Error Rate (WER)	$\leq 2\%$	$\leq 1\%$

Warning

Clinical Standard: These thresholds are intentionally stringent because annotations may inform clinical

diagnosis (e.g., stuttering severity assessment) and therapeutic decisions. Data falling below minimum thresholds must not be used for clinical applications without re-annotation and adjudication.

Note

Interpreting Cohen's Kappa:

- $\kappa < 0.40$: Poor agreement, requires retraining
- $0.40 \leq \kappa < 0.60$: Moderate agreement, review guidelines
- $0.60 \leq \kappa < 0.75$: Substantial agreement, acceptable with monitoring
- $0.75 \leq \kappa < 0.90$: Excellent agreement, meets quality standards
- $\kappa \geq 0.90$: Near-perfect agreement

12.5.3 Calculating Agreement

Step 1: Align annotations

Both annotators' outputs must be aligned at the dysfluency event level. For each potential dysfluency location:

- Record whether each annotator marked a dysfluency (presence)
- If both marked it, record the type assigned by each (type agreement)
- If both marked it, record the severity assigned by each (severity agreement)

Step 2: Build contingency tables

Example: Dysfluency Presence Contingency Table

	Annotator B: Yes	Annotator B: No	Total
Annotator A: Yes	45	5	50
Annotator A: No	3	147	150
Total	48	152	200

Observed agreement: $(45 + 147)/200 = 96\%$

Cohen's $\kappa = \frac{P_o - P_e}{1 - P_e}$ where P_o = observed agreement, P_e = expected by chance

Step 3: Calculate metrics

Use statistical software (R, Python, SPSS) to compute:

- Cohen's κ for categorical variables
- Weighted κ (linear or quadratic weights) for ordinal variables (severity)
- Intraclass Correlation Coefficient (ICC) for continuous measures

12.5.4 Adjudication Process

When annotators disagree, follow this adjudication protocol:

1. **Independent review**: Each annotator re-examines their annotation without seeing the other's work
2. **Joint review**: Both annotators listen to the audio segment together
3. **Discussion**: Annotators discuss the disagreement, referencing these guidelines
4. **Resolution**: If agreement is reached, record the consensus annotation
5. **Escalation**: If no agreement, a senior reviewer or project lead makes the final decision
6. **Documentation**: Record the disagreement type and resolution for training purposes

Table 37: Common Disagreement Types and Resolution Strategies

Disagreement	Common Cause	Resolution Strategy
Presence vs. absence	Subtle dysfluency or normal variation	Re-listen at reduced speed; consider speaker's baseline
Type confusion (R vs. P)	Ambiguous acoustic signal	Focus on whether sound repeats or stretches continuously
Block audibility	Quiet tension sounds	Use headphones; increase volume; check waveform
Severity rating	Boundary placement	Re-measure using waveform; take average if within 1 level
Stuttering vs. normal	Natural repetition for emphasis	Consider context and speaker intent

12.5.5 Annotator Training and Calibration

1. **Initial training**: All annotators complete training on 50+ pre-annotated samples
2. **Certification**: Annotators must achieve $\kappa \geq 0.75$ on a test set before independent work
3. **Calibration sessions**: Weekly meetings to review difficult cases and maintain consistency
4. **Drift monitoring**: Monthly IRR checks to detect annotator drift over time
5. **Retraining**: Required if κ drops below minimum thresholds

12.5.6 Reporting IRR Results

All published datasets and research using SAML annotations should report:

- Number of annotators and their qualifications
- Percentage of data dual-annotated
- Cohen's κ (or Fleiss' κ for >2 annotators) for each annotation dimension
- Weighted κ or ICC for severity ratings
- Adjudication rate (percentage of items requiring adjudication)
- Description of adjudication process used

12.6 Batch Annotation Protocol

Large-scale annotation projects require standardized procedures for file management, work distribution, and quality control. This section provides guidelines for batch annotation workflows.

12.6.1 File Naming Convention

Use consistent naming to enable automated processing and tracking:

Table 38: Recommended File Naming Convention

Component	Format
Audio files	`[project]_[speaker]_[session]_[chunk].wav`
Annotation files	`[project]_[speaker]_[session]_[chunk]_[annotator].saml`
Adjudicated files	`[project]_[speaker]_[session]_[chunk]_gold.saml`

Example:

- Audio: `stutter_P001_S02_C015.wav`
- Annotation: `stutter_P001_S02_C015_A03.saml`
- Gold standard: `stutter_P001_S02_C015_gold.saml`

Note

Naming rules:

- Use lowercase letters and underscores only

- Zero-pad numbers for consistent sorting (P001, not P1)
- Include annotator ID to track who annotated each file
- Never include spaces or special characters

12.6.2 Directory Structure

Organize project files using this structure:

```
project_root/
 raw_audio/           # Original recordings
    speaker_001/
        session_01/
 chunked_audio/       # Segmented audio files
    speaker_001/
        session_01/
 annotations/         # Annotator work products
    annotator_01/
    annotator_02/
    ...
 gold_standard/       # Adjudicated annotations
 irr_samples/         # Dual-annotation subset
 training/            # Training materials
    examples/
    answer_keys/
 exports/             # Final deliverables
    xml/
    shorthand/
 documentation/       # Project docs, guidelines
```

12.6.3 Work Distribution

Guidelines for assigning work to annotators:

1. **Random assignment**: Use random or pseudo-random assignment to avoid systematic bias
2. **Speaker distribution**: Each annotator should work on multiple speakers (avoid one annotator per speaker)
3. **Batch sizes**: Assign 1–2 hours of audio per annotator per day to maintain quality
4. **Deadline buffers**: Build in time for quality review and re-annotation

Table 39: Recommended Work Distribution

Parameter	Recommendation
Audio per annotator per day	60–120 minutes
Chunks per batch	50–100
IRR overlap	20% of total (different annotators)
Training set per project	10–20 gold-standard clips
Re-annotation buffer	10% of timeline

12.6.4 Quality Sampling Strategy

Not all annotations can be reviewed. Use stratified sampling:

1. **Random sample**: 5% of all annotations for spot-checking
2. **First-batch review**: 100% review of each annotator's first batch
3. **Flagged items**: Review any items annotators mark as uncertain
4. **Outlier detection**: Review annotations with unusually high or low dysfluency counts
5. **New annotators**: 50% review for first 2 weeks, then reduce to standard rate

12.6.5 Progress Tracking

Maintain a tracking spreadsheet with these columns:

Table 40: Progress Tracking Template

Column	Description
File ID	Unique identifier for audio chunk
Speaker	Speaker ID
Session	Recording session
Duration	Audio length in seconds
Assigned To	Annotator ID
Date Assigned	Assignment date
Date Completed	Completion date
Status	Pending / In Progress / Complete / QA / Approved
IRR Sample	Yes/No (is this file dual-annotated?)
QA Result	Pass / Fail / Needs Review
Notes	Any issues or comments

12.6.6 Handoff Checklist

Before submitting completed annotations:

- ☐ All assigned files are complete (no missing annotations)
- ☐ File names follow project convention
- ☐ Files are in correct directory structure
- ☐ Self-review checklist completed for each file (Section 12)
- ☐ XML validation passed (if using full XML format)
- ☐ Uncertain items flagged for review
- ☐ Progress tracker updated
- ☐ No duplicate files in submission
- ☐ Metadata matches audio files
- ☐ Backup copy retained until QA approval

Warning

Before Final Submission: Verify that your annotations do not contain any personally identifiable information (PII) beyond what is in the original audio. If you notice PII in the audio (names, addresses, etc.), flag the file for review rather than including it in the transcript.

A
ASR and TTS Consumption Patterns (Non-Normative)

This appendix provides guidance for machine learning practitioners on how to consume SAML-annotated data in ASR and TTS pipelines. This section is **non-normative** and represents recommended practices.

A.1 ASR Training Considerations

A.1.1 Dysfluency Handling Strategies

When training ASR models on stuttered speech, consider these approaches:

Table 41: ASR Dysfluency Handling Strategies

Strategy	Description
Include all	Train on verbatim transcripts including all dysfluencies. Model learns to transcribe stuttered speech as-is.
Fluent targets	Use only the target/intended words as training labels. Disfluencies in audio align to fluent text.
Tagged output	Train model to output dysfluency tags as special tokens (e.g., `[REP]`, `[BLOCK]`).
Multi-task	Jointly train for transcription and dysfluency detection as separate output heads.

A.1.2 Alignment Recommendations

- **Frame-level alignment**: Use forced alignment tools to map dysfluency boundaries to audio frames
- **Phoneme alignment**: For fine-grained analysis, align at phoneme level using tools like Montreal Forced Aligner
- **Boundary precision**: The `duration` attribute (when present) provides millisecond-level timing
- **Chunking alignment**: Chunk boundaries should align with silence regions in audio

A.1.3 Data Augmentation

Stuttered speech datasets are often small. Consider:

- Speed perturbation (0.9x–1.1x) to vary dysfluency durations
- Mixing stuttered and fluent data with appropriate sampling weights
- Synthetic dysfluency insertion using TTS (see below)

Warning

Clinical Validity of Synthetic Dysfluencies: TTS-generated dysfluencies should be used **only** for expanding training data diversity in machine learning applications. Synthetic dysfluencies should **not** be considered equivalent to natural stuttered speech for:

- Clinical assessment or diagnosis
- Research conclusions about stuttering characteristics
- Therapeutic applications

Any dataset using synthetic augmentation **must** clearly disclose this in its documentation, and synthetic samples should be flagged or segregated from natural speech samples.

A.2 TTS Synthesis Considerations

A.2.1 Rendering Disfluencies

When using SAML to control TTS output:

Table 42: TTS Rendering Guidance

Element	Rendering Approach
`<saml:repetition>`	Synthesize each `<unit>` with brief pauses between
`<saml:prolongation>`	Extend phoneme duration based on severity
`<saml:block>`	Insert silence (inaudible) or tense phonation (audible)

A.2.2 Severity to Duration Mapping for TTS

Table 43: Recommended TTS Duration Mapping

Severity	Suggested TTS Duration
0	250–500 ms
1	500–1000 ms
2	1000–2000 ms
3	2000–3000 ms
4–8	Scale proportionally

A.3 Data Export Formats

For ML pipelines, SAML data can be exported to:

- **JSON**: For Python/JavaScript ML frameworks
- **CSV**: For tabular analysis (one row per dysfluency event)
- **TextGrid**: For Praat-compatible phonetic analysis
- **CTM**: For NIST-style scoring and alignment

A.4 SSML Compatibility Notes

SAML extends SSML with custom elements for stuttering annotation. This section documents compatibility considerations for implementations.

A.4.1 SAML vs SSML Feature Comparison

Table 44: SAML Elements and SSML Compatibility

SAML Element	SSML Support	Notes
`<saml:repetition>`	× SAML-specific	No SSML equivalent
`<saml:prolongation>`	× SAML-specific	No SSML equivalent
`<saml:block>`	× SAML-specific	No SSML equivalent
`<saml:nonspeech>`	× SAML-specific	SSML uses `<audio>` for sounds
`<saml:unintelligible>`	× SAML-specific	No SSML equivalent
`<saml:letter>`	✓Maps to	`<say-as interpret-as="characters">`
Interjections (plain text)	✓Compatible	Different purpose than SSML speechcons

A.4.2 TTS Engine Fallback Behavior

When SAML-annotated documents are processed by TTS engines that do not support SAML extensions:

- **Graceful degradation**: Compliant XML parsers should ignore unknown namespace elements
- **Non-speech sounds**: Use SSML `<audio>` elements with sound files as fallback
- **Dysfluency rendering**: Custom TTS implementations needed for naturalistic stuttered speech

A.4.3 Platform-Specific Considerations

Platform	Non-Speech Sound Support
W3C SSML Standard	No native support for non-speech sounds
Amazon Alexa/Polly	Via `<audio>` tag or custom "speechcons"
Google Cloud TTS	Via `<audio>` tag only
Microsoft Azure	Via `<audio>` tag or SSML extensions

A.4.4 Interjections: SAML vs SSML Usage

SAML and SSML handle interjections differently:

- **SAML (annotation)**: Interjections are written as plain text for ASR training. They represent what was actually spoken.
- **SSML (synthesis)**: The `<say-as interpret-as="interjection">` element triggers expressive TTS rendering of words like "wow" or "oops."

These are complementary use cases: SAML annotates recorded speech, while SSML controls synthesized speech.

B

Complete Annotated Sample

```
<?xml version="1.0" encoding="UTF-8"?>
<speak version="1.1"
       xmlns="http://www.w3.org/2001/10/synthesis"
       xmlns:saml="https://shunyalabs.ai/2026/saml"
       xml:lang="en-US">

    <saml:metadata>
        <saml:file id="CONV_2024_001"
                   duration="12300"
                   task="conversation"/>
        <saml:speaker id="SPK_042"
                      age-group="adult"
                      severity="moderate"/>
    </saml:metadata>

    <p id="speaker1">
        So I <saml:repetition type="incomplete-syllable" severity="1">
            <saml:unit>w</saml:unit>
            <saml:unit>w</saml:unit>
        </saml:repetition> was going to the store yesterday,
        um, and I <saml:prolongation sound="s" severity="2"
                                     position="initial">
            saw
        </saml:prolongation> my
        <saml:block type="inaudible" sound="consonant"
                    severity="1"/> friend.

        He said, <saml:block type="audible" sound="vowel"
                             severity="1"/>
        Are you coming to the
        <saml:repetition type="syllable" severity="2">
            <saml:unit>pa</saml:unit>
            <saml:unit>pa</saml:unit>
            <saml:unit>pa</saml:unit>
        </saml:repetition> party tonight?

        And I said, <saml:repetition type="incomplete-syllable" severity="1">
            <saml:unit>y</saml:unit>
            <saml:unit>y</saml:unit>
        </saml:repetition> yeah, I'll
        <saml:repetition type="incomplete-syllable" severity="1">
            <saml:unit>b</saml:unit>
            <saml:unit>b</saml:unit>
        </saml:repetition> be there.
    </p>

</speak>
```

C
Shorthand to XML Conversion Reference

C.1 Repetition Conversion

Shorthand	type Attribute
`[R:isyl n]`	`type="incomplete-syllable"`
`[R:syl n]`	`type="syllable"`
`[R:mix n]`	`type="mixed"`

Note

The explicit type marker is required. Legacy shorthand using `[R n]` without a type marker, or using deprecated markers `[R:snd n]` or `[R:wrd n]`, should be flagged for review during conversion. Legacy `:snd` markers should be converted to `type="incomplete-syllable"`, and legacy `:wrd` markers should be converted to `type="syllable"`.

C.2 Block Tag Conversion

Shorthand	type	sound
`[IBC n]`	`inaudible`	`consonant`
`[ABC n]`	`audible`	`consonant`
`[IBV n]`	`inaudible`	`vowel`
`[ABV n]`	`audible`	`vowel`

D
Element Hierarchy and Allowed Parents

Table 45: SAML Element Hierarchy

Element	Allowed Parent(s)	Children	Content
`saml:metadata`	`speak`	`file`, `speaker`	Empty
`saml:file`	`metadata`	None	Empty
`saml:speaker`	`metadata`	None	Empty
`saml:repetition`	`speak`, `p`, `s`	`unit` (2+)	Empty
`saml:unit`	`repetition`	None	Text (grapheme)
`saml:prolongation`	`speak`, `p`, `s`	None	Text (word)
`saml:block`	`speak`, `p`, `s`	None	Empty
`saml:unintelligible`	`speak`, `p`, `s`	None	Empty
`saml:overlap`	`speak`, `p`, `s`	None	Empty
`saml:letter`	`speak`, `p`, `s`	None	Text (letter)
`saml:nonspeech`	`speak`, `p`, `s`	None	Empty
`saml:background`	`speak`, `p`, `s`	None	Empty

Note

SSML Integration: All SAML dysfluency elements can appear within standard SSML elements such as `<prosody>`, `<emphasis>`, and `<voice>`. The `saml:metadata` element should appear only once, at the

beginning of the `<speak>` element. The `saml:background` element can appear anywhere to indicate environmental sounds.

E

Glossary

Background Sounds
Environmental sounds present in a recording that are not produced by the speaker being transcribed. These **should be annotated** using the `<saml:background>` element. Categories include:

- **Traffic/Transportation**: traffic, siren, horn, aircraft, train
- **Environmental**: wind, rain, thunder, birds, animals, water
- **Human Activity**: crowd, conversation, children, footsteps
- **Indoor/Household**: hvac, appliance, tv, music, phone, door
- **Technical/Equipment**: static, hum, echo, mic-noise

Use the `level` attribute to indicate intensity: `soft`, `moderate`, `loud`, or `overwhelming`. See Section 10.5 for detailed guidance and examples.

Block
A dysfluency where airflow or voicing stops completely.

Continuant
A speech sound that can be prolonged (fricatives, nasals, approximants, vowels).

Disfluency
Normal, typical interruptions in the flow of speech that occur in all speakers, including interjections (um, uh), phrase repetitions, multisyllabic word repetitions, and revisions. These are **not** tagged in SAML as they do not represent stuttering behaviors. Compare with *Dysfluency*.

Dysfluency
Stuttering-specific speech interruptions that are involuntary and associated with the speech disorder of stuttering. Includes incomplete syllable repetitions (ISR), syllable repetitions (SR), mixed repetitions, prolongations, and blocks. These **are** tagged in SAML. Compare with *Disfluency*.

Grapheme
A written letter or letter combination representing a speech sound.

Incomplete Syllable Repetition (ISR)
A repetition type where the repeated unit does not form a complete syllable. This includes single sound repetitions (e.g., "b-b-baby") and partial syllable repetitions (e.g., "do-do-dog"). Varying lengths of incomplete syllables (e.g., "d-d-do-do-dog") are also classified as ISR, not as mixed repetition.

Interjections
Filler words or sounds inserted during speech. These are normal disfluencies and should **not** be tagged; write them as plain text only. See detailed categories below.

Namespace
An XML mechanism for distinguishing elements from different vocabularies.

Non-speech Sounds
Sounds produced by the speaker that are not linguistic content. These are marked using the `<saml:nonspeech>` element. See detailed categories below.

Prolongation
A dysfluency where a sound is stretched longer than normal.

Repetition
A dysfluency where an incomplete syllable, syllable, or monosyllabic word is repeated involuntarily. Classified as ISR (incomplete-syllable), SR (syllable), or mixed.

SSML
Speech Synthesis Markup Language; W3C standard for controlling TTS. SAML extends SSML with custom elements for stuttering annotation.

Syllable Repetition (SR)
A repetition type where complete syllables or monosyllabic words are repeated (e.g., "ba-ba-baby", "dog-dog-dog").

XSD
XML Schema Definition; formal specification of XML document structure.

E.1 Detailed Interjection Reference

Interjections are **normal disfluencies** and should be written as **plain text only**; they are **NOT** tagged with any SAML dysfluency elements.

E.1.1 Hesitation Fillers

Interjection	Spelling	Notes
Schwa hesitation	uh	Most common hesitation sound
Nasal hesitation	um	Hesitation with closed lips
Thinking sound	er	British English variant
Pause filler	erm	British English variant
Mild hesitation	eh	Can also indicate questioning

E.1.2 Discourse Markers

Interjection	Example Usage	Notes
like	"I was, like, going home"	Very common in casual speech
you know	"It was, you know, difficult"	Seeking listener agreement
I mean	"I mean, it's fine"	Clarification/correction
well	"Well, I think so"	Topic transition
so	"So, what happened was..."	Narrative continuation
anyway	"Anyway, moving on..."	Topic change
basically	"Basically, it works like this"	Simplification marker
actually	"Actually, I changed my mind"	Correction/contrast
right	"Right, so we need to..."	Confirmation seeking
okay/OK	"Okay, let's start"	Acknowledgment/transition
literally	"I literally can't even"	Emphasis (often informal)
totally	"I totally agree"	Strong agreement/emphasis
kind of/kinda	"It's kind of weird"	Hedging/softening
sort of/sorta	"I sort of understand"	Hedging/softening
I guess	"I guess that's fine"	Uncertainty/reluctant agreement
I think	"I think we should go"	Opinion hedging
I suppose	"I suppose so"	Reluctant agreement
apparently	"Apparently, it's closed"	Reported information
honestly	"Honestly, I don't know"	Sincerity marker
seriously	"Seriously?"	Disbelief/emphasis

E.1.3 Emotional/Reactive Interjections

Interjection	Meaning	Notes
oh	Realization/surprise	Write as plain text
ah	Understanding/realization	Write as plain text
aha	Sudden understanding	Write as plain text
ahem	Attention-getting/clearing throat	Write as plain text
aww	Sympathy/endearment	Write as plain text
duh	Obvious statement	Write as plain text
eww/ew	Disgust	Write as plain text
hmph	Disapproval/skepticism	Write as plain text
meh	Indifference	Write as plain text
ooh	Interest/excitement	Write as plain text
oops	Mistake acknowledgment	Write as plain text
ouch/ow	Pain expression	Write as plain text
phew	Relief	Write as plain text
psst	Getting attention quietly	Write as plain text
sheesh	Exasperation	Write as plain text
shh/shush	Request for silence	Write as plain text
tsk tsk	Disapproval	Write as plain text
ugh	Frustration/disgust	Write as plain text
uh oh	Anticipation of problem	Write as plain text
whoa	Surprise/slow down	Write as plain text
wow	Surprise/amazement	Write as plain text
yay	Celebration	Write as plain text
yikes	Alarm/concern	Write as plain text
yippee	Excitement	Write as plain text
yuck	Disgust	Write as plain text
huh	Confusion/questioning	Write as plain text

E.1.4 Agreement/Acknowledgment Sounds

Interjection	Spelling	Notes
Agreement hum	mm-hmm	Affirmative
Thinking hum	hmm	Contemplation
Disagreement	mm-mm	Negative (with intonation)
Acknowledgment	uh-huh	Affirmative listening
Negative	uh-uh	Disagreement
Casual yes	yeah	Informal affirmative
Casual no	nah	Informal negative

E.1.5 Onomatopoeia in Speech

These sounds may appear when speakers vocalize actions or sensations:

Sound	Meaning/Context	Notes
achoo	Sneezing sound	Write as interjection if vocalized
brr	Cold sensation	Write as plain text
ha ha/haha	Laughter (spelled out)	Use `<saml:nonspeech type="laugh">` for actual laughter
blah blah	Dismissive/placeholder	Write as plain text
tick tock	Time passing	Write as plain text
ding	Alert/realization	Write as plain text
boom	Emphasis/impact	Write as plain text
whoosh	Speed/movement	Write as plain text

Note

Onomatopoeia vs Non-Speech Sounds: If a speaker *says* "achoo" as a word, write it as plain text. If they actually *sneeze*, use `<saml:nonspeech type="other" description="sneeze"/>`.

Note

Regional and Dialect Variations: The spellings above reflect American English conventions. British English speakers may use different variants:

- American "uh" = British "er"
- American "um" = British "erm"
- "eh" may indicate questioning in American English but is more common as a filler in Canadian English
- "blimey", "crikey", "bloody hell" are British exclamations

Use the spelling that best represents what was spoken. Consistency within a corpus is more important than adhering to a specific variant.

Warning

Critical Rule: All interjections listed above are **normal disfluencies** and must be written as **plain text only**. Do **NOT** tag them with `<saml:repetition>` or any other dysfluency element.

Correct: `Um, I think we should, uh, go to the store.`

Incorrect: `<saml:repetition>Um</saml:repetition>, I think...`

E.1.6 Interjection Annotation Examples

The following examples show how interjections appear in properly annotated transcripts:

```
<!-- Interjections written as plain text -->
Um, I think we should go.

<!-- Mixed with dysfluency annotation -->
So, uh, I <saml:repetition type="incomplete-syllable" severity="1">
    <saml:unit>w</saml:unit>
    <saml:unit>w</saml:unit>
</saml:repetition> was going to the store.

<!-- Discourse markers as plain text -->
I mean, it was, like, really difficult, you know?

<!-- Agreement sounds -->
Yeah, mm-hmm, I understand.
```

E.2 Detailed Non-Speech Sound Reference

Non-speech sounds produced by the speaker are marked using the `<saml:nonspeech>` element.

E.2.1 Standard Non-Speech Types (Schema-Defined)

Type Value	Description	SAML Tag
laugh	Laughter of any kind	`<saml:nonspeech type="laugh"/>`
cough	Coughing sound	`<saml:nonspeech type="cough"/>`
sigh	Audible exhale expressing emotion	`<saml:nonspeech type="sigh"/>`
cry	Crying/sobbing sounds	`<saml:nonspeech type="cry"/>`
clear-throat	Throat clearing	`<saml:nonspeech type="clear-throat"/>`
sniff	Sniffing sound (nasal)	`<saml:nonspeech type="sniff"/>`
inhale	Audible breath in	`<saml:nonspeech type="inhale"/>`
exhale	Audible breath out	`<saml:nonspeech type="exhale"/>`
other	Any other non-speech sound	`<saml:nonspeech type="other" description="..."/>`

E.2.2 Additional Non-Speech Sounds (use type="other")

Physiological Sounds:

- Sneeze: `description="sneeze"`
- Hiccup: `description="hiccup"`
- Yawn: `description="yawn"`
- Burp/Belch: `description="burp"`
- Swallow: `description="swallow"`

Mouth/Oral Sounds:

- Lip smack: `description="lip-smack"`
- Tongue click: `description="tongue-click"`
- Teeth click: `description="teeth-click"`
- Whistle: `description="whistle"`

Emotional/Expressive Sounds:

- Gasp: `description="gasp"`
- Groan: `description="groan"`
- Whimper: `description="whimper"`
- Scream/Shriek: `description="scream"`
- Chuckle/Giggle: Use `type="laugh"` or `description="chuckle"`

Effort/Exertion Sounds:

- Grunt: `description="grunt"`
- Huff: `description="huff"`
- Pant: `description="pant"`

Other Vocalizations:

- Tutting/teeth sucking: `description="tut"` or `description="teeth-suck"`
- Sharp intake of breath: `description="sharp-breath"`
- Humming: `description="hum"`
- Singing fragment: `description="singing"`
- Clapping: `description="clap"`
- Finger snap: `description="snap"`
- Blowing (e.g., blowing nose): `description="blow"`

- Spitting: `description="spit"`
- Gargling/gurgling: `description="gargle"`

E.2.3 Sound Annotation Decision Guide

Sound Type	Annotate?	Element to Use
Speaker-produced sounds (cough, laugh, etc.)	**Yes**	`<saml:nonspeech>`
Background sounds (traffic, music, weather)	**Yes**	`<saml:background>`
Equipment noise (mic bump, static, hum)	**Yes**	`<saml:background>` type="static/hum/mic-noise"
Other speaker in conversation	**No**	Tag in their own turn using `<p>`
Nearby conversation (not in dialogue)	**Yes**	`<saml:background>` type="conversation"

Note

Key Distinction:

- `<saml:nonspeech>` = sounds **produced by the speaker** (coughs, laughs, sighs, etc.)
- `<saml:background>` = sounds **from the environment** (traffic, music, weather, other people, equipment)

E.2.4 Non-Speech Sound XML Examples

The following examples show how non-speech sounds are annotated in context:

```
<!-- Speaker clears throat before speaking -->
<saml:nonspeech type="clear-throat"/> Good morning everyone.

<!-- Laughter during speech -->
I just <saml:nonspeech type="laugh"/> couldn't believe it.

<!-- Crying while speaking -->
I just <saml:nonspeech type="cry"/> wanted to tell her
<saml:nonspeech type="inhale"/> that I loved her.

<!-- Using "other" type for sounds not in standard list -->
And then <saml:nonspeech type="other" description="gasp"/>
I saw it was gone.

<!-- Multiple non-speech sounds -->
<saml:nonspeech type="sniff"/> I'm sorry
<saml:nonspeech type="cry"/> this is hard to talk about.

<!-- Non-speech sound with dysfluency -->
<saml:nonspeech type="clear-throat"/>
I <saml:repetition type="incomplete-syllable" severity="1">
    <saml:unit>w</saml:unit>
    <saml:unit>w</saml:unit>
</saml:repetition> want to say something.
```

E.2.5 Shorthand Notation for Non-Speech Sounds

Shorthand	SAML Element
[laughs]	`<saml:nonspeech type="laugh"/>`
[coughs]	`<saml:nonspeech type="cough"/>`
[sighs]	`<saml:nonspeech type="sigh"/>`
[cries]	`<saml:nonspeech type="cry"/>`
[clears throat]	`<saml:nonspeech type="clear-throat"/>`
[sniffs]	`<saml:nonspeech type="sniff"/>`
[inhales]	`<saml:nonspeech type="inhale"/>`
[exhales]	`<saml:nonspeech type="exhale"/>`
[gasps]	`<saml:nonspeech type="other" description="gasp"/>`
[groans]	`<saml:nonspeech type="other" description="groan"/>`

Note

Cross-Reference: For guidance on background sounds annotation (environmental sounds, equipment noise, etc.), see Section 10.5 (Background Sound Annotation).

F
Element Quick Reference

Table 46: SAML Element Quick Reference

Element	Required Attributes	Purpose
`saml:repetition`	type, severity	Mark repeated sounds/syllables/words
`saml:unit`	(none)	Container for repeated segment
`saml:prolongation`	sound, severity	Mark stretched sounds
`saml:block`	type, sound, severity	Mark speech stoppages
`saml:unintelligible`	type	Mark unclear speech
`saml:overlap`	(none)	Mark overlapping speakers
`saml:letter`	(none)	Mark spelled-out letters
`saml:nonspeech`	type	Mark non-speech sounds (speaker-produced)
`saml:background`	type	Mark background sounds (environmental)
`saml:metadata`	(none)	Container for file/speaker info

G
Severity Scale Reference

Table 47: Severity Scale Quick Reference

Rating	Duration
0	< 0.5 seconds
1	≥ 0.5 to < 1.0 seconds
2	≥ 1.0 to < 2.0 seconds
3	≥ 2.0 to < 3.0 seconds
4	≥ 3.0 to < 5.0 seconds
5	≥ 5.0 to < 10.0 seconds
6	≥ 10.0 to < 30.0 seconds
7	≥ 30.0 to < 60.0 seconds
8	≥ 60.0 seconds

Note

Boundary values are assigned to the higher severity rating.

H
Changelog and Revision History

This appendix documents the version history of the SAML specification. Implementers should reference this section to understand changes between versions and ensure compatibility.

H.1 Version 1.0 (Current)

Table 48: Version 1.0 Release Summary

Attribute	Value
Version	1.0
Status	Community Draft
Release Date	January 2026
Base Standard	W3C SSML 1.1
Namespace	`https://shunyalabs.ai/2026/saml`
Schema	`https://shunyalabs.ai/2026/saml/saml.xsd`
Specification	`https://shunyalabs.ai/saml`

H.1.1 Initial Release Features

This is the initial public release of SAML. Key features include:

- **Core dysfluency elements**: `<saml:repetition>`, `<saml:prolongation>`, `<saml:block>`
- **Repetition classification**: Three-type system (ISR, SR, Mixed)
- **Block classification**: Four subtypes (IBC, ABC, IBV, ABV)
- **Severity scale**: 0–8 scale based on SSI-4 duration thresholds
- **Supporting elements**: `<saml:unintelligible>`, `<saml:nonspeech>`, `<saml:background>`, `<saml:overlap>`, `<saml:letter>`, `<saml:metadata>`
- **Shorthand notation**: Compact annotation format for rapid transcription
- **Reference tools**: Shorthand-to-XML converter with validation support

H.1.2 Available Tools and Resources

The following tools and resources are available as part of the SAML v1.0 release:

Table 49: SAML v1.0 Tools and Resources

Resource	Description	Location
XML Schema (XSD)	Complete W3C-compliant schema for validation	`saml.xsd`
Shorthand Converter	Python CLI/library for shorthand-to-XML conversion	`converter.py`
Sample Annotations	Valid SAML documents demonstrating all features	`schema/tests/`
Specification PDF	This document in PDF format	`saml-specification.pdf`

All resources are available at https://shunyalabs.ai/saml. The converter supports:

- Single file and batch conversion
- Schema validation (with `lxml`)
- Metadata generation (file ID, speaker info, task type)
- Dysfluency statistics reporting
- Multi-speaker transcript handling

H.1.3 Classification System

SAML v1.0 uses a streamlined repetition classification system:

Table 50: SAML v1.0 Repetition Classification

Type	XML Value	Coverage
Incomplete Syllable Repetition (ISR)	`incomplete-syllable`	Single sounds and partial syllables
Syllable Repetition (SR)	`syllable`	Complete syllables and monosyllabic words
Mixed	`mixed`	ISR + SR combination only

Note

Design Decision: Earlier drafts included separate "sound" and "word" repetition types. These were consolidated into ISR and SR respectively to reduce redundancy and improve annotator consistency. Monosyllabic word repetitions (e.g., "I-I-I") are classified as SR because the repeated unit equals a complete syllable.

H.2 Migration and Compatibility

H.2.1 Upgrading from Pre-release Versions

If you have annotations from pre-release drafts, apply these transformations:

Table 51: Pre-release to v1.0 Migration

Pre-release	SAML v1.0
`type="sound"`	`type="incomplete-syllable"`
`type="word"`	`type="syllable"`
`:snd` (shorthand)	`:isyl`
`:wrd` (shorthand)	`:syl`
Namespace year 2025	Namespace year 2026

H.2.2 SSML Compatibility

SAML documents are valid SSML documents with namespace extensions. Standard SSML processors will:

- Parse the document without errors (namespaced elements are ignored)
- Extract plain text content for synthesis
- Not process dysfluency annotations (requires SAML-aware processor)

H.3 Future Versions

Planned enhancements for future versions (not yet implemented):

- **Language-specific annexes**: Adaptations for non-English languages
- **Secondary behaviors**: Annotations for physical concomitants (eye blinks, head movements)
- **Emotional state markers**: Annotations for speaker affect during dysfluencies
- **Covert stuttering**: Support for annotating internal stuttering behaviors

H.4 Version History Template

Future versions will be documented using this format:

```
## Version X.Y (Date)

### Breaking Changes
- [List any backwards-incompatible changes]

### New Features
- [List new elements, attributes, or capabilities]

### Deprecations
- [List features scheduled for removal]

### Bug Fixes
- [List corrections to specification text]

### Migration Guide
- [Step-by-step upgrade instructions]
```

I
Practice Exercises

This appendix provides practice exercises for annotator training. Work through these exercises before annotating real data, then check your answers against the provided solutions.

I.1 Exercise Set 1: Beginner (Single Dysfluency Types)

These exercises contain single, clear dysfluency events. Focus on identifying the type and assigning severity.

I.1.1 Exercise 1.1: Repetitions

Audio transcript (duration of dysfluency events shown in brackets):

> "I want to [0.8s] b-b-b-baby. Can you [1.2s] p-please help me?"

Tasks:

1. Identify both dysfluency events
2. Determine the repetition type for each (ISR, SR, or Mixed)
3. Assign severity ratings based on duration
4. Write the shorthand annotation

Your answer:

```
(write your annotation here)
```

I.1.2 Exercise 1.2: Prolongations

Audio transcript:

"I [1.5s] ssssaw the [0.4s] mmman."

Tasks:

1. Identify both prolongation events
2. Note the prolonged sound and position (initial/medial/final)
3. Assign severity ratings
4. Write the shorthand annotation

Your answer:

```
(write your annotation here)
```

I.1.3 Exercise 1.3: Blocks

Audio transcript:

"Let's [2.1s silent pause] go to the [1.8s audible tension] store."

Tasks:

1. Determine block type for each (IBC, ABC, IBV, ABV)
2. Assign severity ratings
3. Write the shorthand annotation

Your answer:

```
(write your annotation here)
```

I.2 Exercise Set 2: Intermediate (Mixed Types and Edge Cases)

These exercises contain multiple dysfluency types and require careful judgment.

I.2.1 Exercise 2.1: Multiple Dysfluency Types

Audio transcript:

"[1.0s silent pause] Um, I-I-I [0.6s] th-think that we should, you know, [2.5s] ggggo."

Tasks:

1. Identify which events are stuttering dysfluencies (tag) vs. normal disfluencies (don't tag)
2. Classify each stuttering event
3. Write the complete annotation

Hints:

- "Um" and "you know" are filler words; do not tag
- "I-I-I" is a monosyllabic word repetition
- Consider whether the initial pause shows signs of struggle

Your answer:

```
(write your annotation here)
```

I.2.2 Exercise 2.2: ISR vs Mixed

Audio transcript:

"I want some [0.9s] t-t-to-to-toffee."

Question: Is this ISR or Mixed? Explain your reasoning.

Your answer:

I.2.3 Exercise 2.3: Intentional vs Involuntary

Classify each as **Tag** (stuttering) or **Don't Tag** (intentional/normal):

1. "No no no, that's not right" (speaker shaking head)
2. "I-I-I don't know" (speaker appears to struggle)
3. "The biiiig one" (emphasizing size)
4. "The [tense pause] big one" (speaker appears stuck)
5. "I went to... I mean I drove to the store"

Your answers:

I.3 Exercise Set 3: Advanced (Complex Scenarios)

These exercises simulate real annotation challenges.

I.3.1 Exercise 3.1: Full Passage Annotation

Audio transcript (approximate durations shown):

> "[0.3s] So, um, [1.2s silent block] yesterday I-I-I [0.7s] went to the, uh, [2.0s] sssstore to get some [0.5s] b-b-bread. The, you know, the [1.5s audible block] cashier was [0.8s] re-re-really nice."

Tasks:

1. Annotate all stuttering dysfluencies
2. Do NOT annotate normal disfluencies (um, uh, you know)
3. Use shorthand notation
4. Check your work against the severity scale

Your answer:

```
(write your annotation here)
```

I.3.2 Exercise 3.2: Ambiguous Cases

For each scenario, state your annotation decision and reasoning:

1. A speaker says "ho-ho-honest" with slight tension. The word "honest" starts with a silent H. What type is this?
2. A speaker repeats "I-I" and then pauses for 2 seconds before continuing. How do you annotate the repetition vs. any potential block?
3. You hear "d-d-do-do-dog" where the "d-d" units are very short (0.1s each) and "do-do" are longer (0.3s each). Total duration is 0.8s. Is this ISR or Mixed?

Your answers:

I.4 Answer Key

I.4.1 Answers for Exercise Set 1

Exercise 1.1:

```
I want to ( [R:isyl 1] b- b- b- ) baby. Can you ( [R:isyl 2] p- ) please help me?
```

Explanation: "b-b-b" is ISR (single consonant, not complete syllable). Duration 0.8s = severity 1. "p-" is ISR. Duration 1.2s = severity 2.

Exercise 1.2:

```
I s( [P 2] s- )aw the m( [P 0] m- )an.
```

Explanation: "sss" prolonged 1.5s = severity 2 (initial position). "mmm" prolonged 0.4s = severity 0 (initial position).

Exercise 1.3:

```
Let's ( [IBC 3] ) go to the ( [ABC 2] ) store.
```

Explanation: First block is silent before consonant /g/ = IBC, 2.1s = severity 3. Second block has audible tension before consonant /s/ = ABC, 1.8s = severity 2.

I.4.2 Answers for Exercise Set 2

Exercise 2.1:

```
( [IBC 1] ) Um, ( [R:syl 1] I- I- I- ) ( [R:isyl 1] th- ) think that we should, you know, g( [P 3] g- )o.
```

Notes:

- Initial 1.0s silent pause: If there's evidence of struggle, tag as IBC (severity 1). If it's just a thinking pause, don't tag.
- "Um" and "you know": NOT tagged (fillers)
- "I-I-I": SR (monosyllabic word), severity 1
- "th-": ISR, severity 1
- "ggg": Prolongation, 2.5s = severity 3

Exercise 2.2:

This is **Mixed** (R:mix). The sequence contains both:

- "t-t-" = incomplete syllable (just the consonant)
- "to-to-" = complete syllable of "toffee"

When you have both ISR and SR units in the same repetition sequence, it's Mixed.

Exercise 2.3:

1. **Don't Tag**: Intentional emphatic repetition
2. **Tag** (R:syl): Involuntary, with struggle
3. **Don't Tag**: Intentional prolongation for emphasis
4. **Tag** (Block): Involuntary stoppage
5. **Don't Tag**: Revision/false start

I.4.3 Answers for Exercise Set 3

Exercise 3.1:

```
So, um, ( [IBC 2] ) yesterday ( [R:syl 1] I- I- I- ) went to the, uh, s( [P 3] s- )tore to get some ( [R:isyl 0] b- b- ) bread. The, you know, the ( [ABC 2] ) cashier was ( [R:syl 1] re- re- ) really nice.
```

Notes:

- 0.3s initial pause: Not tagged (too short, no mention of struggle)
- "um", "uh", "you know": NOT tagged

- 1.2s silent block before "yesterday" = IBC severity 2
- "I-I-I" = SR (monosyllabic), severity 1
- "sss" 2.0s = P severity 3
- "b-b" 0.5s = ISR severity 0 (under 0.5s)
- 1.5s audible block = ABC severity 2
- "re-re-" = SR (complete syllable of "really"), severity 1

Exercise 3.2 Answers:

1. **SR** (syllable). The "ho" in "ho-ho-honest" is a complete syllable even though "h" is silent. The phonetic onset is vowel /o/, so if there were a block it would be IBV/ABV, but since it's a repetition of "ho-" it's SR.
2. Annotate separately: The "I-I" repetition gets one annotation (R:syl), and if the 2-second pause shows signs of struggle/blocking, it gets a separate block annotation. If the pause is just thinking time, don't tag the pause.
3. **ISR** (incomplete-syllable). Even though the units vary in length (some "d", some "do"), they are all representing the same incomplete unit. The key is whether you have *both* incomplete AND complete syllables. Here, "do" is not a complete syllable of "dog"; the complete first syllable would be "dog" itself (it's monosyllabic). So all units are incomplete attempts = ISR.

I.5 Self-Scoring Rubric

Use this rubric to evaluate your practice exercise performance:

Table 52: Self-Scoring Rubric

Skill	Points	Scoring Guide
Dysfluency detection	2 pts each	2 = correctly identified; 1 = partially identified; 0 = missed
Type classification	2 pts each	2 = correct type; 1 = close (e.g., ISR vs Mixed confusion); 0 = wrong
Severity rating	1 pt each	1 = correct or within ±1; 0 = off by 2+
Non-tagging decisions	1 pt each	1 = correctly not tagged; 0 = incorrectly tagged

Readiness thresholds:

- **≥90%**: Ready to begin annotating with periodic review
- **80–89%**: Review incorrect answers, redo exercises
- **<80%**: Review Sections 2, 6, and 8 before reattempting

J

Frequently Asked Questions

This appendix addresses common questions from annotators and implementers.

J.1 General Questions

Q: What is the difference between "dysfluency" and "disfluency"?

A: In this specification, **dysfluency** refers specifically to stuttering-related speech interruptions (repetitions, prolongations, blocks). **Disfluency** refers to normal speech interruptions that everyone produces (um, uh, false starts, phrase repetitions). Only dysfluencies are tagged in SAML.

Q: Do I need to know SSML to use SAML?

A: For annotation work, no. You can use the shorthand notation without any XML knowledge. For software development, basic SSML/XML understanding is helpful but the specification provides complete examples.

Q: Can I use SAML for languages other than English?
A: Yes. The core framework is language-agnostic. See Section 1.5.2 for adaptation guidance. You'll need to document language-specific conventions for filler words, syllabification, etc.

Q: What audio software should I use?
A: Audacity (free) or Praat (free) are recommended. Any software that displays waveforms and allows millisecond-precision measurement will work.

J.2 Classification Questions

Q: How do I classify "I-I-I want"?
A: This is **Syllable Repetition (SR)**. "I" is a monosyllabic word, and when repeated, the unit equals a complete syllable.

Q: Is "um" a dysfluency?
A: No. "Um", "uh", "er", "like", "you know" are normal filler words (disfluencies). Do not tag them. Write them in the transcript as plain text.

Q: What if someone repeats a two-syllable word like "maybe-maybe"?
A: Do not tag. Multi-syllable word repetitions are considered normal disfluency, not stuttering, even if they occur in a person who stutters.

Q: How do I handle "d-d-do-do-dog" with varying unit lengths?
A: If all units are incomplete syllables (even if varying in length), classify as **ISR**. Only use **Mixed** when you have both incomplete ("d-") AND complete (e.g., "dog-" in a hypothetical "d-d-dog-dog-dog") syllable units.

Q: What's the difference between IBC and ABC?
A: **IBC** (Inaudible Block Consonant) = complete silence before a consonant-initial word. **ABC** (Audible Block Consonant) = you can hear tension, struggle, or partial sounds before a consonant-initial word. If in doubt, increase volume and check the waveform.

Q: How do I classify a block before "honest"?
A: Based on **phonetic onset**, not spelling. "Honest" begins with vowel sound /o/ (the "h" is silent), so it would be IBV or ABV depending on audibility.

J.3 Severity Questions

Q: How do I handle boundary values like exactly 1.0 seconds?
A: Boundary values go to the **higher** rating. Exactly 1.0s = severity 2, exactly 2.0s = severity 3, etc.

Q: What if I can't measure precisely?
A: Estimate to the nearest 0.5 seconds. When in doubt, use the **lower** severity rating. Consistency is more important than precision.

Q: Should I include the target word in repetition duration?
A: No. Measure from the start of the first repeated unit to the end of the last repeated unit, **excluding** the final target word.

J.4 Annotation Process Questions

Q: Do I use full XML or shorthand?
A: Use shorthand for faster annotation during transcription. Convert to full XML only if required by your project's tooling. Section 11 and Appendix C provide conversion guidance.

Q: How do I handle overlapping speech?
A: Use the `<saml:overlap>` element. Annotate dysfluencies in the primary speaker's turn and note overlapping content separately.

Q: What if I hear a dysfluency but can't identify the word?
A: Use `<saml:unintelligible>` with `type="guess"` if you can estimate the word, or `type="unknown"` if you cannot.

Q: How long should my audio chunks be?
A: Target 10 seconds (acceptable range: 5–15 seconds). Never break chunks in the middle of a dysfluency event. See Section 1.8 for detailed chunking guidelines.

J.5 Edge Cases

Q: What if someone stutters on a filler word like "um"?
A: This is rare but can happen. If there's clear involuntary repetition or blocking on "um" itself (e.g., "u-u-u-um"), you may tag it. Use clinical judgment.

Q: How do I annotate covert stuttering (internal struggle with no audible dysfluency)?
A: Currently, SAML does not have elements for covert stuttering. If the speaker reports internal struggle but produces fluent speech, there is nothing to annotate. Future versions may address this.

Q: What about secondary behaviors (eye blinks, head movements)?
A: SAML v1.0 focuses on audible speech patterns. Secondary behaviors are not currently annotated. Video annotation protocols may be developed separately.

Q: How do I handle speech in a very noisy environment?
A: Use `<saml:background>` to document noise. If dysfluencies are obscured by noise and you cannot reliably identify them, note this limitation in metadata or comments. Do not guess at dysfluencies you cannot hear clearly.

K
Tool Configuration Guides

This appendix provides setup instructions for common audio analysis tools used in SAML annotation.

K.1 Audacity Configuration

Audacity is a free, cross-platform audio editor suitable for SAML annotation.

K.1.1 Recommended Settings

1. **Download**: Get Audacity from https://www.audacityteam.org/
2. **Sample rate display**: Edit → Preferences → Quality → Default Sample Rate: 44100 Hz
3. **Time display**: Click the time display (bottom left) and select "hh:mm:ss + milliseconds"
4. **Spectrogram view** (optional): Track dropdown → Spectrogram. Useful for visualizing prolongations.

K.1.2 Creating SAML Annotation Labels

1. **Add label track**: Tracks → Add New → Label Track
2. **Create label**: Select audio region, then Ctrl+B (Cmd+B on Mac)
3. **Label naming convention**:
   - Use shorthand notation in labels: `R:isyl 1`, `P 2`, `IBC 3`
   - One label per dysfluency event
4. **Export labels**: File → Export → Export Labels (creates tab-delimited file)

K.1.3 Measuring Duration

1. Zoom in until individual sounds are visible (Ctrl+1 to zoom in)
2. Click at start of dysfluency, drag to end
3. Read duration from selection display at bottom of window
4. For precision, use View → Zoom → Zoom to Selection

K.2 Praat Configuration

Praat is a specialized phonetics tool with powerful analysis features.

K.2.1 Installation and Setup

1. **Download**: Get Praat from https://www.fon.hum.uva.nl/praat/
2. **Open sound**: Open → Read from file
3. **View**: Select sound, click "View & Edit"

K.2.2 Creating TextGrid for SAML

TextGrids are Praat's annotation format. Create a multi-tier TextGrid for SAML:

1. Select sound object

2. Click "Annotate" → "To TextGrid"
3. Create tiers:
   - **Tier 1** (interval): "transcript": full text transcription
   - **Tier 2** (interval): "dysfluency": dysfluency annotations
   - **Tier 3** (point): "severity": severity ratings
4. Click "OK"

K.2.3 Annotation Workflow

1. Open sound and TextGrid together (View & Edit)
2. On "transcript" tier: type verbatim transcription
3. On "dysfluency" tier: select region, type shorthand (e.g., `R:isyl`)
4. On "severity" tier: click at dysfluency midpoint, type severity number
5. Save TextGrid: Save → Save as text file

K.2.4 Measuring Duration in Praat

1. Click and drag to select region
2. Duration appears in the info bar (top of editor window)
3. For precise boundaries, zoom with scroll wheel or Ctrl+I/Ctrl+O
4. Use spectrogram view (Show spectrogram) to identify sound boundaries

K.3 Recommended Display Settings

For optimal dysfluency identification:

Table 53: Recommended Audio Display Settings

Setting	Audacity	Praat
View mode	Waveform + Spectrogram	Show spectrogram
Zoom level	50–200 ms visible	0.5–2 s window
Spectrogram range	0–8000 Hz	0–5000 Hz
Time precision	milliseconds	Show duration

Note

Tip: Spectrogram view is especially helpful for:

- Identifying prolongation boundaries (sustained formant patterns)
- Detecting audible blocks (low-intensity noise patterns)
- Distinguishing repetition units (repeated spectral patterns)